Rational Resonances and Microscopic Natural–Logarithmic Profiles for the Liouville Function

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Abstract

Let $\lambda(n) = (-1)^{\Omega(n)}$. For $X \in \mathbb{N}$ and $u \in \mathbb{R}$, define

$$C_X(u) := \sum_{\substack{1 \le n \le X \\ \lfloor e^u n \rfloor \ge 1}} \lambda(n)\lambda(\lfloor e^u n \rfloor). \tag{0.1}$$

For reduced rational dilations $e^u = p/q$, we isolate exactly the unique proportional residue class. Assuming the standard fixed two-point affine Chowla conjecture, this yields the natural microscopic profile

$$\frac{1}{X} C_X\left(\log \frac{p}{q} + \frac{c}{X}\right) \longrightarrow \begin{cases} 0, & c < 0, \\ \lambda(pq)/q, & 0 \le c \le q/p, \\ \lambda(pq)/(pc), & c > q/p. \end{cases} \tag{0.2}$$

The logarithmically weighted analogue has an unconditional one-sided microscopic limit at the same X^{-1} scale. At integer resonances we prove, without Chowla input, a right-hand natural-average majority window $0 < c < 2/m$ and a logarithmic mesoscopic majority theorem whose elementary majority mechanism has threshold exponent $1/2$: every sequence $c_X = X^{\theta+o(1)}$ with $\theta < 1/2$ forces non-cancellation at the corresponding right-hand scale $c_X/X = X^{-1+\theta+o(1)}$. Thus natural and logarithmic weighting exhibit different microscopic scales, while integer resonances are singled out by unconditional majority rigidity. In this expanded version the two-point input is

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supplied by the ordinary two-point theorem of OpenAI (2026), which makes the natural profile unconditional with the rate $(\log X)^{-\kappa}$. We introduce the k-point resonance family, prove its exact decomposition into proportionality classes and its closed-form profile, extend the plateaus, the majority theorem and the left-hand identity to every k, and show that the level-three family at the resonances $(2, 3)$ is, for every large X, the ordinary three-point sum $\sum_{M\equiv 3\,(6)} \lambda(M)\lambda(M+1)\lambda(M+3)$. For the bias method behind the two-point theorem we prove that its centering step cannot be extended to $k \geq 3$ with the short-interval input it uses, that local U^k uniformity suffices for that step, and that a quantitative local Gowers uniformity hypothesis together with a k-linear divisor-graph hypothesis yields k-point Chowla in progressions and the profile of the whole family.

Keywords: Liouville function, Chowla conjecture, Beatty sequence, rational resonance, logarithmic average, microscopic profile

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1. Introduction

The Liouville function $\lambda(n) = (-1)^{\Omega(n)}$ is completely multiplicative. The fixed two-point Chowla conjecture predicts natural-average cancellation of λ on non-proportional affine forms. Tao proved the corresponding logarithmically averaged two-point theorem [6], while Matomäki–Radziwiłł–Tao proved an averaged natural form [5]. Arithmetic functions on Beatty sequences were studied by Abercrombie [1] and Abercrombie–Banks–Shparlinski [2].

For irrational α with first slope equal to 1, logarithmic cancellation of $\lambda(n)\lambda(\lfloor \alpha n \rfloor)$ was proved by Crnčević, Hernández, Rizk, Sereesuchart and R. Tao [4]. For a general irrational ratio of slopes, logarithmic two-point cancellation is due to Teräväinen and Walker [7, Theorem 1.2]; they moreover prove that in the rationally dependent case the logarithmic limit exists and characterize when it vanishes, answering the two-point case of a conjecture of Frantzikinakis [3]. The statement of Teräväinen–Walker, Theorem 1.2(3), gives existence and a zero/non-zero criterion in the rationally dependent case, but it does not state a closed-form value for the present specialization. Here $\alpha_1 = 1$, $\alpha_2 = p/q$ and $r = q/p$. Their vanishing criterion cannot hold eventually, because for every ℓ divisible by q,

$$\ell = \frac{q}{p} \left\lfloor \frac{p\ell}{q} \right\rfloor . \tag{1.1}$$

Thus their theorem implies non-vanishing in this case. The explicit value $\lambda(pq)/q$ is obtained here by isolating the proportional residue class and

applying Tao's logarithmic two-point theorem to the finitely many non-proportional affine classes. The new object is the singular X-dependent perturbation

$$u_X = \log(p/q) + c/X, \tag{1.2}$$

together with its natural microscopic profile and the unconditional majority phenomena that appear at integer resonances.

The structure is transparent. At a reduced rational dilation p/q, the residue class $n \equiv 0 \pmod{q}$ is the unique proportional affine class; all other residue classes are non-proportional. The natural main term is therefore the sign $\lambda(pq)$ times the size of this proportional block, while the remaining pieces require affine Chowla cancellation.

Expanded version. Sections 7–9 are new. The remainder of the residue-class decomposition is a finite set of fixed binary affine correlations, and the ordinary two-point Chowla theorem of [10, Theorem 1.1, p. 3] controls exactly these; Section 7 records the resulting unconditional profile with a rate, and shows that at the trivial resonance $p = q = 1$, $-1 < c < 0$, the profile is literally the shift-one Chowla sum. Section 8 constructs the k-point family $C_X^{(k)}$, proves its exact decomposition (Theorem 8.3), the criterion for unconditional closure (Proposition 8.8), the plateau, majority and left-hand identities at every level (Theorem 8.10), the identification of level three at $(2, 3)$ with an ordinary three-point Chowla sum (Corollary 8.11), and the necessity theorem (Theorem 8.12). Section 9 examines the bias method of [10, Part II] for $k \geq 3$ points: a barrier for its centering step (Propositions 9.1 and 9.2), an unconditional bound of the k-point centering error by local Gowers norms (Proposition 9.5), and a conditional k-point theorem (Theorem 9.10, Corollary 9.11) whose two hypotheses are stated explicitly and are not proved here.

2. Affine input and logarithmic normalization

Conjecture 2.1 (Standard fixed two-point affine Chowla)**.** *Let $a_1, a_2 \in \mathbb{N}$ and $b_1, b_2 \in \mathbb{Z}$ be fixed, with the two affine forms positive for all sufficiently large n and $a_1 b_2 - a_2 b_1 \neq 0$. Then*

$$\sum_{\substack{n \leq Y \\ a_1 n + b_1 \geq 1,\ a_2 n + b_2 \geq 1}} \lambda(a_1 n + b_1)\lambda(a_2 n + b_2) = o(Y). \tag{2.1}$$

No uniform or quantitative strengthening of this conjecture is used.

Throughout the paper, $X \in \mathbb{N}$ and all limits $X \to \infty$ are taken through positive integers. For $u \in \mathbb{R}$ we write

$$C_X(u) := \sum_{\substack{1 \le n \le X \\ \lfloor e^u n \rfloor \ge 1}} \lambda(n)\lambda(\lfloor e^u n \rfloor). \tag{2.2}$$

Lemma 2.2 (Macroscopic subintervals)**.** *Assume Conjecture 2.1 for a fixed pair of affine forms. If $0 \le A_X \le B_X$, $A_X/X \to A$ and $B_X/X \to B$ with $0 \le A \le B < \infty$, then*

$$\sum_{\substack{A_X < n \le B_X \\ a_1 n + b_1 \ge 1,\ a_2 n + b_2 \ge 1}} \lambda(a_1 n + b_1)\lambda(a_2 n + b_2) = o(X).$$

Proof. Let $S(Y)$ be the sum in (2.1). The conjecture says that for every $\varepsilon > 0$ there is C_ε with $|S(Y)| \le \varepsilon Y + C_\varepsilon$ for all $Y \ge 0$. The sum in the lemma is $S(B_X) - S(A_X)$, so it is at most $\varepsilon(A_X + B_X) + 2C_\varepsilon \le 3\varepsilon BX$ for large X (and $\le \varepsilon X + 2C_\varepsilon$ if $B = 0$). This covers $A = 0$ and $A = B$. □

Let

$$H_X := \sum_{n \le X} \frac{1}{n} \tag{2.3}$$

and define

$$C_X^{\log}(u) := \frac{1}{H_X} \sum_{\substack{1 \le n \le X \\ \lfloor e^u n \rfloor \ge 1}} \frac{\lambda(n)\lambda(\lfloor e^u n \rfloor)}{n}. \tag{2.4}$$

Thus $|C_X^{\log}(u)| \le 1$.

Lemma 2.3 (Affine logarithmic cancellation with residue-class weight)**.** *Let $A_1, A_2, Q \in \mathbb{N}$ and $B_1, B_2, A \in \mathbb{Z}$ be fixed. Assume that the affine forms $A_1 k + B_1$ and $A_2 k + B_2$ are positive for all sufficiently large k and are non-proportional:*

$$A_1 B_2 - A_2 B_1 \ne 0. \tag{2.5}$$

If $Qk + A > 0$ for all sufficiently large k, then

$$\sum_{\substack{1 \le k \le K \\ A_1 k + B_1 \ge 1,\ A_2 k + B_2 \ge 1 \\ Qk + A \ge 1}} \frac{\lambda(A_1 k + B_1)\lambda(A_2 k + B_2)}{Qk + A} = o(\log K). \tag{2.6}$$

Proof. Tao's logarithmically averaged two-point theorem [6, Theorem 1.2] states that for fixed $a_1, a_2 \in \mathbb{N}$ and $b_1, b_2 \in \mathbb{Z}$ with $a_1 b_2 - a_2 b_1 \neq 0$, and for any $1 \leq \omega(K) \leq K$ with $\omega(K) \to \infty$,

$$\sum_{K/\omega(K)<k\leq K} \frac{\lambda(a_1 k + b_1)\lambda(a_2 k + b_2)}{k} = o(\log \omega(K)). \tag{2.7}$$

Apply this with $(a_i, b_i) = (A_i, B_i)$ and $\omega(K) = K$. Deleting or restoring finitely many initial terms changes the sum by $O(1)$, so

$$\sum_{k\leq K} \frac{\lambda(A_1 k + B_1)\lambda(A_2 k + B_2)}{k} = o(\log K). \tag{2.8}$$

The residue-class weight satisfies

$$\frac{1}{Qk + A} = \frac{1}{Q}\,\frac{1}{k} + O_{Q,A}(k^{-2}). \tag{2.9}$$

Since the Liouville product has modulus one, the total contribution of the $O(k^{-2})$ error is $O(1)$. Combining (2.8) and (2.9) proves (2.6). □

Remark 2.4 (Status of Conjecture 2.1). For non-negative intercepts, Conjecture 2.1 is now [10, Theorem 1.1, p. 3], an unrefereed preprint, with a power-of-logarithm saving at every real cutoff; negative intercepts reduce to that case by a fixed shift (Lemma 7.2). Sections 3–6 are kept in their original conditional formulation; the unconditional statements are collected in Section 7.

Remark 2.5 (Logical status of the results). The results of this paper fall into four classes.

(A) *Conditional on Conjecture 2.1:* Lemma 2.2, Theorem 3.2, Theorem 5.1 and Corollary 6.1, as stated in Sections 3–6.

(B) *Unconditional, by exact identities or classical and logarithmic inputs:* Lemma 2.3 and Theorem 4.1 (via Tao's logarithmic two-point theorem), Proposition 3.1, Theorems 6.2 and 6.4, Proposition 6.8, Lemma 8.1, Theorem 8.3, Lemma 8.4, Theorem 8.7(1), Theorem 8.10, Theorem 8.12 (an implication), Theorem 8.13 (via Tao–Teräväinen), and Propositions 9.1, 9.2, Lemma 9.4, Proposition 9.5, Lemma 9.9 in Section 9.

(C) *Unconditional only through Theorem 7.1, i.e. through the unrefereed preprint [10]:* Lemma 7.2, Theorem 7.3, Corollary 7.4, Theorem 8.7(2) and Proposition 8.8. If [10] is withdrawn or corrected, these return to class (A).

(D) *Conditional on hypotheses for $k \geq 3$:* Theorem 8.7(3) and Corollary 8.11 (on $\mathrm{Ch}_{k'}$), Theorem 9.7 (on LU_k), Theorem 9.10 and Corollary 9.11 (on LU_k and G_k). Besides these hypotheses, Theorems 9.7 and 9.10 also use intermediate estimates of [10] (the rough-number bound $\varpi \ll B^{-1+\varepsilon}\log B$, the divisor-interval and dilation lemmas, and the order of limits), so they too depend on that unrefereed preprint.

No other result of the paper uses [10]: the results of classes (A) and (B) do not use it, and the literature entries of Theorem 7.1 and Subsection 9.1 only quote it.

3. Exact rational decomposition

Proposition 3.1 (Exact residue-class decomposition)**.** *Let $(p,q) = 1$ and $X \in \mathbb{N}$. Then*

$$C_X\left(\log\frac{p}{q}\right) = \lambda(pq)\left\lfloor\frac{X}{q}\right\rfloor + \sum_{a=1}^{q-1}\sum_{\substack{k\geq 0\\ 1\leq qk+a\leq X\\ pk+\lfloor pa/q\rfloor\geq 1}} \lambda(qk+a)\lambda\left(pk+\left\lfloor\frac{pa}{q}\right\rfloor\right). \tag{3.1}$$

For every $a \neq 0$ the two affine forms in the inner sum are non-proportional.

Proof. Write $n = qk + a$, $0 \leq a < q$. Then

$$\left\lfloor\frac{p}{q}(qk+a)\right\rfloor = pk + \left\lfloor\frac{pa}{q}\right\rfloor. \tag{3.2}$$

For $a = 0$ and $k \geq 1$, complete multiplicativity gives $\lambda(qk)\lambda(pk) = \lambda(pq)$. For $a \neq 0$, proportionality would require $q\lfloor pa/q\rfloor = pa$, impossible because $(p,q) = 1$ and $q \nmid a$. □

Theorem 3.2 (Fixed rational natural resonance)**.** *Assume Conjecture 2.1. Then*

$$\frac{1}{X}C_X\left(\log\frac{p}{q}\right) \longrightarrow \frac{\lambda(pq)}{q}. \tag{3.3}$$

For $q = 1$ the exact identity

$$C_X(\log p) = \lambda(p)X \tag{3.4}$$

holds for every $X \in \mathbb{N}$.

Proof. The proportional class in Proposition 3.1 contributes $\lambda(pq)\lfloor X/q \rfloor$. Every other residue class is a fixed non-proportional affine correlation and is $o(X)$ by Conjecture 2.1. For $q = 1$ there are no remaining classes. □

4. Unconditional logarithmic microscopic step

Theorem 4.1 (Logarithmic microscopic step)**.** *Let $(p, q) = 1$ and let $c \in \mathbb{R}$ be fixed. Then*

$$C_X^{\log}\left(\log \frac{p}{q} + \frac{c}{X}\right) \longrightarrow \frac{\lambda(pq)}{q}\, \mathbf{1}_{\{c \geq 0\}} \tag{4.1}$$

unconditionally.

Proof. Fix $\varepsilon \in (0, 1)$ and set

$$Y = X^{1-\varepsilon}. \tag{4.2}$$

Write $n = qk + a$, $0 \leq a < q$. Uniformly for $n \leq Y$,

$$\frac{p}{q}(e^{c/X} - 1)n = O_{p,q,c}(X^{-\varepsilon}). \tag{4.3}$$

If $a = 0$, then $\{pa/q\} = 0$. When $q = 1$ there are no nonzero residue classes. If $a \neq 0$, coprimality gives

$$\left\{\frac{pa}{q}\right\} \in \left[\frac{1}{q}, 1 - \frac{1}{q}\right]. \tag{4.4}$$

Hence, for all sufficiently large X, the perturbation in (4.3) cannot change the floor on any $a \neq 0$ residue class.

Suppose first that $c \geq 0$. On $n \leq Y$ the perturbed correlation agrees exactly with the fixed rational dilation. The proportional class $a = 0$ contributes

$$\frac{\lambda(pq)}{H_X} \sum_{k \leq Y/q} \frac{1}{qk} = \frac{H_Y}{H_X} \frac{\lambda(pq)}{q} + o(1). \tag{4.5}$$

For each $a \neq 0$, the two fixed affine forms are non-proportional, so Lemma 2.3, with weight $(qk+a)^{-1}$, shows that its entire contribution on $n \leq Y$ is $o(H_Y)$ and therefore $o(1)$ after division by H_X. The tail $Y < n \leq X$ has normalized harmonic mass

$$\frac{H_X - H_Y}{H_X} = \varepsilon + o(1). \tag{4.6}$$

Consequently,

$$\left| C_X^{\log}\left(\log \frac{p}{q} + \frac{c}{X}\right) - \frac{H_Y}{H_X}\frac{\lambda(pq)}{q} \right| \leq \frac{H_X - H_Y}{H_X} + o(1). \tag{4.7}$$

Since $H_Y/H_X \to 1-\varepsilon$, first letting $X \to \infty$ and then $\varepsilon \downarrow 0$ gives the branch $c \geq 0$. Teräväinen–Walker's Theorem 1.2(3) already gives the rationally dependent zero/non-zero alternative and implies non-vanishing in the present homogeneous rational case; the closed-form value $\lambda(pq)/q$ is computed here from the proportional residue class [7, Theorem 1.2(3)].

Now suppose that $c < 0$. On the class $a = 0$ and the range $n = qk \leq Y$, (4.3) gives

$$\left\lfloor pe^{c/X}k \right\rfloor = pk - 1 \tag{4.8}$$

for every $k \geq 1$ once X is large, except for the possible initial term for which the target equals zero and is excluded by definition. To apply Tao in the usual non-negative-intercept formulation, write $k = \ell + 1$:

$$qk = q\ell + q, \qquad pk - 1 = p\ell + (p-1). \tag{4.9}$$

If $p = 1$, the single term with second value zero is omitted; the remaining terms may be reindexed once more. The two forms are non-proportional. Every $a \neq 0$ class remains non-proportional by (4.4). Lemma 2.3 therefore makes the whole initial range $o(H_Y)$, while the tail is bounded by (4.6). Letting $X \to \infty$ and then $\varepsilon \downarrow 0$ gives zero. □

5. Conditional natural microscopic profile

Theorem 5.1 (Rational natural microscopic profile)**.** *Assume Conjecture 2.1.*

Let $(p,q) = 1$ and fix $c \in \mathbb{R}$. Then

$$\frac{1}{X} C_X\left(\log \frac{p}{q} + \frac{c}{X}\right) \longrightarrow \Psi_{p,q}(c), \tag{5.1}$$

where

$$\Psi_{p,q}(c) = \begin{cases} 0, & c < 0, \\ \dfrac{\lambda(pq)}{q}, & 0 \le c \le q/p, \\ \dfrac{\lambda(pq)}{pc}, & c > q/p. \end{cases} \tag{5.2}$$

For $c > 0$,

$$\Psi_{p,q}(c) = \lambda(pq) \min\left(\frac{1}{q}, \frac{1}{pc}\right). \tag{5.3}$$

Proof. Put

$$\delta_X := \frac{p}{q}(e^{c/X} - 1) \tag{5.4}$$

and write $n = qk + a$, $0 \le a < q$. Then

$$\left\lfloor e^{\log(p/q)+c/X}(qk + a) \right\rfloor = pk + \left\lfloor \frac{pa}{q} + \delta_X(qk + a) \right\rfloor. \tag{5.5}$$

Assume first $c > 0$. For an integer r, the block on which the final floor in (5.5) equals $\lfloor pa/q \rfloor + r$ is

$$r \le \left\{ \frac{pa}{q} \right\} + \delta_X(qk + a) < r + 1. \tag{5.6}$$

Equivalently,

$$\frac{r - \{pa/q\}}{Xp(e^{c/X} - 1)} - \frac{a}{qX} \le \frac{k}{X} < \frac{r + 1 - \{pa/q\}}{Xp(e^{c/X} - 1)} - \frac{a}{qX}. \tag{5.7}$$

This is written for $c > 0$. For $c < 0$ the factor $Xp(e^{c/X} - 1)$ is negative, dividing by it reverses the inequalities, and (5.7) holds with its two endpoints interchanged; nothing below depends on their order. Since $Xp(e^{c/X} - 1) \to pc$, both endpoints converge. After intersection with $0 \le k/X \le 1/q + O(X^{-1})$, each block has endpoints converging to fixed limits. If the limiting endpoints are distinct, the block is macroscopic; if they coincide, its length is $o(X)$; otherwise it is empty. For fixed p, q, c, the perturbation in (5.5) is uniformly bounded for $0 \le n \le X$, so the number of integers r that occur is bounded in terms of p, q, c and is independent of X.

On such a block the two affine forms are

$$qk + a, \qquad pk + \left\lfloor \frac{pa}{q} \right\rfloor + r. \tag{5.8}$$

For $a \neq 0$, proportionality would require

$$qr = pa - q \left\lfloor \frac{pa}{q} \right\rfloor = pa \bmod q \in \{1, \ldots, q-1\}, \tag{5.9}$$

which is impossible. For $a = 0$, proportionality occurs only for $r = 0$. Thus every block except $(a, r) = (0, 0)$ is $o(X)$ by Lemma 2.2.

The unique proportional block is

$$n = qk, \qquad p(e^{c/X} - 1)k < 1. \tag{5.10}$$

Its cardinality is

$$R_X(c) = \min\left(\left\lfloor \frac{X}{q} \right\rfloor, \left\lceil \frac{1}{p(e^{c/X} - 1)} \right\rceil - 1 \right), \tag{5.11}$$

up to the harmless endpoint convention. Hence

$$\frac{R_X(c)}{X} \longrightarrow \min\left(\frac{1}{q}, \frac{1}{pc} \right), \tag{5.12}$$

and every term in this block equals $\lambda(pq)$.

If $c < 0$, the same block decomposition has negative corrections. For $a = 0$ the correction is a nonzero negative integer for every $k \geq 1$, so no proportional block remains. For $a \neq 0$, proportionality would again imply $qr = pa \bmod q$, impossible. Only finitely many negative correction integers occur for fixed p, q, c. For each such correction, the target affine form can fail to be positive only for $O_{p,q,c}(1)$ initial values of k; these terms are omitted by the definition of C_X. All remaining blocks are $o(X)$.

At $c = 0$, Theorem 3.2 applies. At the boundary $pc = q$,

$$\frac{pX}{q}(e^{c/X} - 1) = 1 + \frac{q}{2pX} + O(X^{-2}), \tag{5.13}$$

so only $O(1)$ terminal terms leave the proportional block. The normalized limit remains $\lambda(pq)/q$. □

6. Integer rigidity and unconditional majority

Corollary 6.1 (Integer profile and exact right-hand plateau)**.** *Assume Conjecture 2.1. For fixed $m \geq 2$,*

$$\frac{1}{X} C_X\Big(\log m + \frac{c}{X}\Big) \longrightarrow \begin{cases} 0, & c < 0, \\ \lambda(m), & 0 \leq c \leq 1/m, \\ \lambda(m)/(mc), & c > 1/m. \end{cases} \tag{6.1}$$

Moreover, for every fixed $0 < c < 1/m$,

$$C_X\Big(\log m + \frac{c}{X}\Big) = \lambda(m) X \tag{6.2}$$

for all sufficiently large X.

Proof. The limit is Theorem 5.1 with $q = 1$. For the exact statement, $mX(e^{c/X} - 1) \to mc < 1$, so eventually no correction occurs for any $n \leq X$. □

Theorem 6.2 (Unconditional right-hand natural majority window)**.** *Let $m \geq 1$ and $0 < c < 2/m$. Then*

$$\liminf_{X \to \infty} \lambda(m) \frac{1}{X} C_X\Big(\log m + \frac{c}{X}\Big) \geq 2 \min\Big(1, \frac{1}{mc}\Big) - 1 > 0. \tag{6.3}$$

Proof. Let

$$I_{0,X} := \{1 \leq n \leq X : m(e^{c/X} - 1)n < 1\}. \tag{6.4}$$

Every summand on $I_{0,X}$ equals $\lambda(m)$, while every remaining term has modulus at most one. Therefore

$$\lambda(m) C_X\Big(\log m + \frac{c}{X}\Big) \geq 2|I_{0,X}| - X. \tag{6.5}$$

Since

$$\frac{|I_{0,X}|}{X} \to \min\Big(1, \frac{1}{mc}\Big), \tag{6.6}$$

the result follows. □

Remark 6.3 (Signed natural majority bound). Theorem 6.2 is a signed lower bound. In particular, absolute non-cancellation follows immediately.

Theorem 6.4 (Unconditional logarithmic mesoscopic majority)**.** *Let* $(p,q) = 1$ *and let* $c_X > 0$ *satisfy*

$$\frac{\log c_X}{\log X} \longrightarrow \theta, \qquad 0 \le \theta < 1. \tag{6.7}$$

Then

$$\liminf_{X\to\infty} \lambda(pq)\, C_X^{\log}\!\left(\log\frac{p}{q} + \frac{c_X}{X}\right) \ge \frac{2(1-\theta)}{q} - 1. \tag{6.8}$$

In particular, when $q = 1$ *and* $\theta < 1/2$*, writing* $m = p$*,*

$$\liminf_{X\to\infty} \lambda(m) C_X^{\log}\!\left(\log m + \frac{c_X}{X}\right) \ge 1 - 2\theta > 0. \tag{6.9}$$

Proof. On the proportional class $n = qk$, the zero-correction condition is

$$p(e^{c_X/X} - 1)k < 1. \tag{6.10}$$

Let K_X be the largest admissible k, truncated by $k \le X/q$. Since $c_X = X^{\theta+o(1)}$ and $\theta < 1$,

$$K_X = X^{1-\theta+o(1)}. \tag{6.11}$$

The normalized harmonic mass of this block is

$$B_X = \frac{1}{H_X} \sum_{k\le K_X} \frac{1}{qk} = \frac{1-\theta}{q} + o(1). \tag{6.12}$$

After multiplication by $\lambda(pq)$, all terms on this block are positive. The complement has total normalized harmonic mass at most $1 - B_X$, so

$$\lambda(pq) C_X^{\log} \ge B_X - (1 - B_X) = 2B_X - 1. \tag{6.13}$$

This proves (6.8). □

Remark 6.5 (Threshold of the elementary logarithmic majority mechanism). For $q = 1$, the lower bound in Theorem 6.4 is positive exactly when $\theta < 1/2$. Thus $1/2$ is the threshold exponent for this elementary majority mechanism, not a claimed transition exponent for the correlation itself. More precisely, every sequence $c_X = X^{\theta+o(1)}$ with $0 \le \theta < 1/2$ is covered, at the displacement $c_X/X = X^{-1+\theta+o(1)}$; in particular this includes $X^{-1/2-\varepsilon+o(1)}$ for every $\varepsilon > 0$. For $q \ge 2$, this elementary majority bound yields no positive lower bound.

Remark 6.6 (Positive fixed logarithmic branch from the majority theorem). For $q = 1$ and $\theta = 0$, Theorem 6.4 gives $\liminf \lambda(m)C_X^{\log} \geq 1$, while $|C_X^{\log}| \leq 1$. Hence

$$C_X^{\log}\Big(\log m + \frac{c_X}{X}\Big) \to \lambda(m) \tag{6.14}$$

for every positive c_X with $\log c_X / \log X \to 0$. No logarithmic Chowla input is needed for this integer positive branch.

Remark 6.7 (The endpoint $c = 1/m$ is not an exact finite plateau). For the integer right-hand layer,

$$Xm(e^{1/(mX)} - 1) = 1 + \frac{1}{2mX} + O(X^{-2}) > 1. \tag{6.15}$$

Thus $c = 1/m$ belongs to the closed limiting plateau. Call c in the *exact finite plateau* if $C_X(\log m + c/X) = \lambda(m)X$ for all sufficiently large X. At $c = 1/m$ only the term $n = X$ changes, and $C_X(\log m + \frac{1}{mX}) = \lambda(m)(X-1) + \lambda(X)\lambda(mX+1)$; this differs from $\lambda(m)X$ whenever $\lambda(X)\lambda(mX+1) = -\lambda(m)$, which happens for infinitely many X because $\sum_{X \leq Y} \lambda(X)\lambda(mX+1) = o(Y)$ (Theorem 7.1). Hence $c = 1/m$ is not in the exact finite plateau. For $m = 2$, $X = 10^6$, the regenerated computation gives $C_X = -999998$.

Proposition 6.8 (Exact left-hand c-independence interval)**.** *Let $m \geq 2$ and fix $-1/m < c < 0$. For all sufficiently large X,*

$$C_X\Big(\log m + \frac{c}{X}\Big) = \sum_{n \leq X} \lambda(n)\lambda(mn - 1). \tag{6.16}$$

This states exact independence with respect to c on that finite interval. At this point Theorem 7.1 is not used, and no asymptotic value of the right-hand sum is asserted here; Corollary 7.4 later bounds it by $O(X(\log X)^{-\kappa})$.

Proof. Put $\beta_X = m(1 - e^{c/X})$. Since $X\beta_X \to m|c| < 1$, eventually $0 < \beta_X n < 1$ for all $n \leq X$. Hence $\lceil \beta_X n \rceil = 1$ and $\lfloor me^{c/X} n \rfloor = mn - 1$. □

Proposition 6.9 (Locally finite step structure)**.** *Fix $m \geq 2$ and $X \in \mathbb{N}$. The map*

$$c \longmapsto C_X\Big(\log m + \frac{c}{X}\Big) \tag{6.17}$$

is locally constant except at explicit crossing values. Every compact c-interval contains only finitely many jumps. For sufficiently negative c, disappearance points where $\lfloor me^{c/X} n \rfloor < 1$ are included among these crossings.

Proof. For each fixed $n \le X$, the integer $\lfloor me^{c/X}n \rfloor$ changes only when the continuous function $me^{c/X}n$ crosses an integer. On a compact c-interval its range is bounded, hence only finitely many crossings occur. There are only finitely many $n \le X$. □

7. Unconditional natural profile

Conjecture 2.1 is used in Sections 5 and 6 only for finitely many fixed pairs of non-proportional affine forms, namely those produced by the residue-class blocks of Proposition 3.1 and of the proof of Theorem 5.1. The following theorem supplies exactly this input, with a rate.

Theorem 7.1 (Ordinary two-point affine Chowla [10, Theorem 1.1, p. 3])**.** *There is an absolute constant $\kappa > 0$ with the following property. Let $a_1, a_2 \ge 1$ and $b_1, b_2 \ge 0$ be fixed integers with $a_1b_2 - a_2b_1 \ne 0$. Then there is $C = C_{a_1,a_2,b_1,b_2}$ such that*

$$\Big|\sum_{n\le X} \lambda(a_1n + b_1)\lambda(a_2n + b_2)\Big| \le C\,\frac{X}{(\log X)^{\kappa}} \qquad (X \ge 3,\ X \in \mathbb{R}). \tag{7.1}$$

The constants in (7.1) may be ineffective, and no uniformity in coefficients growing with X is asserted in [10, p. 3]; none is used below.

Lemma 7.2 (Macroscopic blocks with a rate)**.** *Let $\alpha_1, \alpha_2 \ge 1$ and $\beta_1, \beta_2 \in \mathbb{Z}$ be fixed with $\alpha_1\beta_2 - \alpha_2\beta_1 \ne 0$. Let $0 \le A \le B < \infty$ be fixed and $A_X = AX + O(1)$, $B_X = BX + O(1)$. Then*

$$\sum_{\substack{A_X<k\le B_X\\ \alpha_1k+\beta_1\ge 1,\ \alpha_2k+\beta_2\ge 1}} \lambda(\alpha_1k + \beta_1)\lambda(\alpha_2k + \beta_2) = O\Big(\frac{X}{(\log X)^{\kappa}}\Big). \tag{7.2}$$

Proof. Choose $k_0 \ge 0$ with $\alpha_ik_0+\beta_i \ge 0$ for $i = 1, 2$ and substitute $k = \ell+k_0$. The forms become $\alpha_i\ell + (\alpha_ik_0 + \beta_i)$, with non-negative intercepts and the same determinant $\alpha_1\beta_2 - \alpha_2\beta_1 \ne 0$. The positivity conditions remove only $O(1)$ terms. The sum is a difference of two prefix sums at the cutoffs A_X and B_X. Theorem 7.1 applies to each cutoff $\asymp X$; when $A = 0$ the lower prefix sum has $A_X = O(1)$ terms and is $O(1)$. □

Theorem 7.3 (Rational natural microscopic profile, unconditional)**.** *Let $(p, q) = 1$ and fix $c \in \mathbb{R}$. Then*

$$\frac{1}{X}\,C_X\Big(\log\frac{p}{q} + \frac{c}{X}\Big) = \Psi_{p,q}(c) + O_{p,q,c}\Big((\log X)^{-\kappa}\Big), \tag{7.3}$$

with $\Psi_{p,q}$ as in (5.2) *and κ the absolute constant of Theorem 7.1.*

Proof. For $c = 0$ use Proposition 3.1: the proportional class contributes $\lambda(pq)\lfloor X/q\rfloor$ and each of the $q-1$ remaining classes is a sum of the shape (7.2) with $A = 0$, $B = 1/q$. Let $c \neq 0$. In the proof of Theorem 5.1 the number of pairs (a, r) is bounded in terms of p, q, c. Since $Xp(e^{c/X} - 1) = pc + O(X^{-1})$, the endpoints in (5.7) are $X\,\frac{r-\{pa/q\}}{pc} + O(1)$; after intersection with $0 \le k \le X/q$ every block is an interval $(A_X, B_X]$ with $A_X = AX + O(1)$, $B_X = BX + O(1)$ for fixed $A \le B$. Every block other than $(a, r) = (0, 0)$ consists of two non-proportional fixed affine forms, so Lemma 7.2 bounds it by $O(X(\log X)^{-\kappa})$. The proportional block has cardinality $R_X(c) = X\min(1/q, 1/(pc))\mathbf{1}_{c>0} + O(1)$, and each of its terms equals $\lambda(pq)$. □

Corollary 7.4 (Unconditional forms of Theorem 3.2, Corollary 6.1 and Proposition 6.8)**.** *Let $(p, q) = 1$ and $m \ge 2$.*

1. $C_X(\log(p/q)) = \lambda(pq)X/q + O(X(\log X)^{-\kappa})$.
2. *The limit in Corollary 6.1 holds unconditionally, with error term $O_{m,c}((\log X)^{-\kappa})$.*
3. *For fixed $-1/m < c < 0$ and all sufficiently large X,*

$$C_X\Big(\log m + \frac{c}{X}\Big) = \sum_{n\le X} \lambda(n)\lambda(mn-1) = O\Big(\frac{X}{(\log X)^{\kappa}}\Big). \tag{7.4}$$

Proof. Items (1) and (2) are the cases $c = 0$ and $q = 1$ of Theorem 7.3. For (3), the identity is Proposition 6.8, and the forms n, $mn-1$ have determinant $-1 \neq 0$, so Lemma 7.2 applies. □

Remark 7.5 (The trivial resonance is binary Chowla). Take $p = q = 1$ and fix $-1 < c < 0$. Since $X(1 - e^{c/X}) \to |c| < 1$, for $X \ge X_0(c)$ one has $0 < n(1 - e^{c/X}) < 1$ for every $n \le X$, hence $\lfloor e^{c/X} n\rfloor = n - 1$, and the term $n = 1$ is omitted by definition. Thus, for every $X \ge X_0(c)$,

$$C_X\Big(\frac{c}{X}\Big) = \sum_{2\le n\le X} \lambda(n-1)\lambda(n), \tag{7.5}$$

while $\Psi_{1,1}(c) = 0$. The profile of Theorem 5.1 at the single point $(p, q, c) = (1, 1, c)$ is therefore, for every large X and not only in the limit, the shift-one Chowla sum. No argument that avoids two-point input can produce Theorem 5.1; the residue-class decomposition isolates the main term exactly, and the remainder is a finite set of fixed binary correlations, which Theorem 7.1 controls.

8. The k-point resonance family

8.1. Definition and exact decomposition

For $k \geq 2$ and $u = (u_1, \dots, u_{k-1}) \in \mathbb{R}^{k-1}$ put

$$\begin{aligned} C_X^{(k)}(u) &:= \sum_{\substack{1 \leq n \leq X \\ \lfloor e^{u_i} n \rfloor \geq 1 \ \forall i}} \lambda(n) \prod_{i=1}^{k-1} \lambda\Big(\lfloor e^{u_i} n \rfloor\Big), \\ C_X^{(k),\log}(u) &:= \frac{1}{H_X} \sum_{\substack{1 \leq n \leq X \\ \lfloor e^{u_i} n \rfloor \geq 1 \ \forall i}} \frac{\lambda(n) \prod_i \lambda(\lfloor e^{u_i} n \rfloor)}{n}. \end{aligned} \tag{8.1}$$

For $k = 2$ these are $C_X(u)$ and $C_X^{\log}(u)$ of Section 2. Fix coprime $p_i, q_i \geq 1$ and $c_i \in \mathbb{R}$ and set

$$u_i = \log \frac{p_i}{q_i} + \frac{c_i}{X}, \qquad L := \operatorname{lcm}(q_1, \dots, q_{k-1}), \qquad \delta_i := \frac{p_i}{q_i}\Big(e^{c_i/X} - 1\Big). \tag{8.2}$$

Write $n = Lm + a$ with $0 \leq a < L$, $m \geq 0$, and

$$\theta_{a,i} := \left\{ \frac{p_i a}{q_i} \right\}, \qquad b_{a,i} := \left\lfloor \frac{p_i a}{q_i} \right\rfloor, \qquad A_i := \frac{p_i L}{q_i} \in \mathbb{N}. \tag{8.3}$$

Lemma 8.1 (Floors on residue classes)**.** *For $n = Lm + a \leq X$,*

$$\lfloor e^{u_i} n \rfloor = A_i m + b_{a,i} + r_i(n), \qquad r_i(n) := \Big\lfloor \theta_{a,i} + \delta_i n \Big\rfloor, \tag{8.4}$$

and $|r_i(n)| \leq \frac{p_i}{q_i} |c_i| e^{|c_i|} + 1$. In particular $r_i(n)$ ranges over a finite set R_i depending only on (p_i, q_i, c_i).

Proof. $e^{u_i} n = \frac{p_i}{q_i}(Lm + a) + \delta_i n = A_i m + b_{a,i} + \theta_{a,i} + \delta_i n$, and $|\delta_i n| \leq \frac{p_i}{q_i} |e^{c_i/X} - 1| X \leq \frac{p_i}{q_i} |c_i| e^{|c_i|}$ for $n \leq X$. □

For $0 \leq a < L$ and $r = (r_1, \dots, r_{k-1}) \in R := \prod_i R_i$ let $I_{a,r}(X)$ be the set of $m \geq 0$ with $1 \leq Lm + a \leq X$, $f_{a,r,i}(m) \geq 1$ for all i, and $r_i(Lm + a) = r_i$ for all i, where

$$f_{a,r,0}(m) := Lm + a, \qquad f_{a,r,i}(m) := A_i m + b_{a,i} + r_i \quad (1 \leq i \leq k-1). \tag{8.5}$$

Since $n \mapsto \theta_{a,i} + \delta_i n$ is affine, $I_{a,r}(X)$ is an interval of integers, and by Lemma 8.1

$$C_X^{(k)}(u) = \sum_{a=0}^{L-1} \sum_{r \in R} \sum_{m \in I_{a,r}(X)} \prod_{i=0}^{k-1} \lambda\Big(f_{a,r,i}(m)\Big) \tag{8.6}$$

exactly, for every X.

Definition 8.2 (Proportionality classes)**.** For fixed (a, r) write $f_{a,r,i}(m) = \mathcal{A}_i m + \mathcal{B}_i$, let $d_i := \gcd(\mathcal{A}_i, |\mathcal{B}_i|)$ (with $\gcd(\mathcal{A}, 0) = \mathcal{A}$), and let $(\alpha_i, \beta_i) := (\mathcal{A}_i, \mathcal{B}_i)/d_i$, a primitive vector with $\alpha_i \geq 1$. Declare $i \sim j$ if $(\alpha_i, \beta_i) = (\alpha_j, \beta_j)$, equivalently $\mathcal{A}_i\mathcal{B}_j = \mathcal{A}_j\mathcal{B}_i$. For a class $\mathcal{C}$ let $g_{\mathcal{C}}(m) := \alpha_{\mathcal{C}} m + \beta_{\mathcal{C}}$ be its primitive form, let $\mathrm{Odd}(a, r)$ be the set of classes of odd cardinality and $k'(a, r) := |\mathrm{Odd}(a, r)|$. The block is *resonant* if $k'(a, r) = 0$. Put $\varepsilon_{a,r} := \lambda(d_0 d_1 \cdots d_{k-1}) \in \{\pm 1\}$.

Theorem 8.3 (Exact decomposition of the k-point family)**.** *For every $X \geq 1$ and every choice* (8.2)*,*

$$C_X^{(k)}(u) = \sum_{\substack{(a,r) \\ k'(a,r)=0}} \varepsilon_{a,r} \left| I_{a,r}(X) \right| + \sum_{\substack{(a,r) \\ k'(a,r)\geq 1}} \varepsilon_{a,r} \sum_{m \in I_{a,r}(X)} \prod_{\mathcal{C} \in \mathrm{Odd}(a,r)} \lambda\Big(g_{\mathcal{C}}(m)\Big). \tag{8.7}$$

The forms $g_{\mathcal{C}}$, $\mathcal{C} \in \mathrm{Odd}(a, r)$, are primitive, pairwise non-proportional and positive on $I_{a,r}(X)$, and $k'(a, r) \equiv k \pmod 2$.

Proof. On $I_{a,r}(X)$ every $f_{a,r,i}(m) \geq 1$, hence $g_{\mathcal{C}}(m) \geq 1$ and $\lambda(f_{a,r,i}(m)) = \lambda(d_i)\lambda(g_{\mathcal{C}}(m))$ by complete multiplicativity. Since $\lambda^2 = 1$,

$$\prod_{i=0}^{k-1} \lambda(f_{a,r,i}(m)) = \lambda\Big(\prod_i d_i\Big) \prod_{\mathcal{C}} \lambda\Big(g_{\mathcal{C}}(m)\Big)^{|\mathcal{C}|} = \varepsilon_{a,r} \prod_{\mathcal{C} \in \mathrm{Odd}(a,r)} \lambda\Big(g_{\mathcal{C}}(m)\Big). \tag{8.8}$$

Distinct classes have distinct primitive vectors and are therefore non-proportional, and $\sum_{\mathcal{C}} |\mathcal{C}| = k$ gives the parity. Substitute into (8.6). □

Lemma 8.4 (Block lengths)**.** *Let $J_{a,r}(c) \subseteq [0, 1]$ be the set of t with $r_i \leq \theta_{a,i} + \frac{p_i c_i}{q_i} t < r_i + 1$ for all i. It is an interval, possibly empty, and $|I_{a,r}(X)| = \frac{X}{L} |J_{a,r}(c)| + O_{p,q,c}(1)$. More precisely, $\{m : \ m \in I_{a,r}(X)\} = (A_X, B_X] \cap \mathbb{Z}$ with $A_X = AX + O(1)$, $B_X = BX + O(1)$ for fixed $A \leq B$.*

Proof. For $n \leq X$, $\delta_i n = \frac{p_i c_i}{q_i} \frac{n}{X}(1 + O(|c_i|/X))$, so the endpoints of $\{Lm/X : m \in I_{a,r}(X)\}$ differ from those of $J_{a,r}(c)$ by $O(1/X)$, and the conditions $f_{a,r,i}(m) \geq 1$ remove $O(1)$ values of m. □

Definition 8.5 (k-point profile)**.**

$$\Psi_{p,q}^{(k)}(c) := \frac{1}{L} \sum_{\substack{(a,r) \\ k'(a,r)=0}} \varepsilon_{a,r} \left| J_{a,r}(c) \right|. \tag{8.9}$$

For odd k there are no resonant blocks, since $k' \equiv k \pmod 2$, and $\Psi^{(k)} \equiv 0$.

Remark 8.6 ($k = 2$). For $k = 2$, $L = q$, and $0 \sim 1$ holds if and only if $q \mid a$ and $r = 0$, that is $(a, r) = (0, 0)$, with $d_0 = q$, $d_1 = p$, $\varepsilon_{0,0} = \lambda(pq)$, and $|J_{0,0}(c)| = \min(1, q/(pc))$ for $c \geq 0$, $|J_{0,0}(c)| = 0$ for $c < 0$. Definition 8.5 recovers $\Psi_{p,q}$ of Theorem 5.1, and Theorem 8.3 recovers the block structure of its proof.

8.2. What each order of Chowla closes

Say that $\mathrm{Ch}_{k'}$ holds if, for every tuple of k' pairwise non-proportional affine forms $\alpha_j m + \beta_j$ ($\alpha_j \geq 1$, $\beta_j \in \mathbb{Z}$), $\sum_{m \leq Y} \prod_j \lambda(\alpha_j m + \beta_j) = o(Y)$, the finitely many m with some form ≤ 0 being omitted.

Theorem 8.7 (Sufficiency)**.** *In* (8.7)*:*

1. *the pieces with $k'(a, r) = 1$ are $O_{p,q,c}(X \exp(-c_0\sqrt{\log X}))$ for an absolute $c_0 > 0$;*
2. *the pieces with $k'(a, r) = 2$ are $O_{p,q,c}(X(\log X)^{-\kappa})$;*
3. *if $\mathrm{Ch}_{k'}$ holds for every $3 \leq k' \leq k$ with $k' \equiv k \pmod 2$, then*

$$\frac{1}{X} C_X^{(k)}(u) \longrightarrow \Psi_{p,q}^{(k)}(c) \qquad (X \to \infty). \tag{8.10}$$

Proof. By Lemma 8.4 each piece is $\varepsilon_{a,r}$ times a sum over $(A_X, B_X]$, a difference of two prefix sums. (1) For one primitive form $\alpha m + \beta$ this is a sum of λ over a fixed progression, bounded by the prime number theorem in arithmetic progressions of fixed modulus. (2) Two non-proportional primitive forms: Lemma 7.2. (3) The pieces with $k' \geq 3$ are $o(X)$ by $\mathrm{Ch}_{k'}$, and Lemma 8.4 gives the resonant part. □

Proposition 8.8 (Exact criterion for unconditional closure)**.** *Fix (p, q, c). If every block (a, r) with $|J_{a,r}(c)| > 0$ has $k'(a, r) \leq 2$, then* (8.10) *holds unconditionally, with error $O((\log X)^{-\kappa})$. Conversely, if some block with $|J_{a,r}(c)| > 0$ has $k'(a, r) \geq 3$, then Theorem 8.7(1),(2) do not cover it, and* (8.10) *at (p, q, c) is equivalent to the statement that the sum over all such blocks is $o(X)$; it does not follow that each block separately satisfies its k'-point estimate.*

Proof. Blocks with $|J_{a,r}(c)| = 0$ have $O(1)$ elements. The others are covered by Theorem 8.7(1),(2) exactly when $k' \leq 2$. □

Example 8.9 ($k = 4$, closed unconditionally)**.** Take $(p_i/q_i) = (2, 3, 6)$ and $c = (1, c_2, 1)$ with $0 < c_2 < \frac{1}{3}$. Then $L = 1$, $a = 0$, $r_2 = 0$ on $[0, 1]$, $r_1 = \lfloor 2t \rfloor$ and $r_3 = \lfloor 6t \rfloor$. The blocks in $t = n/X$, with the proportionality classes of the forms $(m,\ 2m + r_1,\ 3m,\ 6m + r_3)$ indexed $0, 1, 2, 3$, are:

t	r	classes	piece
$[0, \frac{1}{6})$	$(0, 0, 0)$	$\{0, 1, 2, 3\}$	resonant, $\varepsilon = \lambda(36) = 1$
$[\frac{1}{6}, \frac{1}{3})$	$(0, 0, 1)$	$\{0, 1, 2\}, \{3\}$	$\lambda(6)\lambda(m)\lambda(6m + 1)$, $k' = 2$
$[\frac{1}{3}, \frac{1}{2})$	$(0, 0, 2)$	$\{0, 1, 2\}, \{3\}$	$\lambda(12)\lambda(m)\lambda(3m + 1)$, $k' = 2$
$[\frac{1}{2}, \frac{2}{3})$	$(1, 0, 3)$	$\{0, 2\}, \{1, 3\}$	resonant, $\varepsilon = \lambda(9) = 1$
$[\frac{2}{3}, \frac{5}{6})$	$(1, 0, 4)$	$\{0, 2\}, \{1\}, \{3\}$	$\lambda(6)\lambda(2m + 1)\lambda(3m + 2)$, $k' = 2$
$[\frac{5}{6}, 1)$	$(1, 0, 5)$	$\{0, 2\}, \{1\}, \{3\}$	$\lambda(3)\lambda(2m + 1)\lambda(6m + 5)$, $k' = 2$

By Proposition 8.8, unconditionally,

$$\frac{1}{X}\, C_X^{(4)}\Big(\log 2 + \frac{1}{X},\ \log 3 + \frac{c_2}{X},\ \log 6 + \frac{1}{X}\Big) = \frac{1}{3} + O\Big((\log X)^{-\kappa}\Big). \tag{8.11}$$

The second plateau comes from the block $(1, 0, 3)$, where the four forms pair into $(m, 3m)$ and $(2m + 1, 6m + 3)$ without all four being proportional; this pairing has no analogue for $k = 2$.

8.3. Plateaus, majority and the left-hand identity at integer resonances

Theorem 8.10 (Integer resonances at level k)**.** *Let $m_1, \dots, m_{k-1} \geq 1$ be integers and $u_i = \log m_i + c_i/X$.*

1. *If $0 < c_i < 1/m_i$ for all i, then for $X \geq X_0(m, c)$*

$$C_X^{(k)}(u) = \lambda(m_1 \cdots m_{k-1}) \sum_{n \leq X} \lambda(n)^k = \begin{cases} \lambda(m_1 \cdots m_{k-1})\lfloor X \rfloor, & k \text{ even}, \\ O\Big(X e^{-c_0\sqrt{\log X}}\Big), & k \text{ odd}. \end{cases} \tag{8.12}$$

2. *If k is even and $c_i > 0$ for all i, then*

$$\liminf_{X \to \infty}\ \lambda(m_1 \cdots m_{k-1}) \frac{C_X^{(k)}(u)}{X}\ \geq\ 2 \min\Big(1, \frac{1}{\max_i m_i c_i}\Big) - 1. \tag{8.13}$$

3. *If $-1/m_i < c_i < 0$ for all i, then for $X \geq X_0(m, c)$*

$$C_X^{(k)}(u) = \sum_{\substack{n \leq X \\ m_i n - 1 \geq 1\ \forall i}} \lambda(n) \prod_{i=1}^{k-1} \lambda(m_i n - 1). \tag{8.14}$$

No Chowla input is used.

Proof. $\lfloor m_i e^{c_i/X} n \rfloor = m_i n + \lfloor m_i (e^{c_i/X} - 1) n \rfloor$ for $c_i > 0$. (1) Since $m_i X (e^{c_i/X} - 1) \to m_i c_i < 1$, eventually the correction vanishes for every $n \le X$, and $\lambda(n) \prod_i \lambda(m_i n) = \lambda(\prod m_i) \lambda(n)^k$; for odd k use the prime number theorem with classical error term. (2) On $n < \min_i \{m_i (e^{c_i/X} - 1)\}^{-1}$ every correction vanishes and every summand equals $\lambda(\prod m_i) \lambda(n)^k = \lambda(\prod m_i)$; the proportion of such $n \le X$ tends to $\min(1, 1/\max_i m_i c_i)$, and every other summand has modulus one. (3) Since $m_i X (1 - e^{c_i/X}) \to m_i |c_i| < 1$, eventually $0 < m_i (1 - e^{c_i/X}) n < 1$ for $n \le X$, so $\lfloor m_i e^{c_i/X} n \rfloor = m_i n - 1$. □

Parts (1) and (2) extend Corollary 6.1 and Theorem 6.2 to every k; part (3) extends Proposition 6.8. If the m_i are pairwise distinct, the k forms $n, m_1 n - 1, \dots, m_{k-1} n - 1$ in (8.14) are pairwise non-proportional, so $k' = k$ and $\Psi^{(k)} = 0$ there. The left-hand plateau is thus, for every large X, a single k-point Chowla sum.

Corollary 8.11 (Level three at $(2, 3)$ is ordinary three-point Chowla)**.** *For every $c = (c_1, c_2) \in (-\frac{1}{2}, 0) \times (-\frac{1}{3}, 0)$ and every $X \ge X_0(c)$,*

$$C_X^{(3)}\Big(\log 2 + \frac{c_1}{X},\ \log 3 + \frac{c_2}{X}\Big) = \sum_{n \le X} \lambda(n) \lambda(2n-1) \lambda(3n-1) = \sum_{\substack{3 \le M \le 6X - 3 \\ M \equiv 3 \pmod 6}} \lambda(M) \lambda(M+1) \lambda(M+3), \tag{8.15}$$

while $\Psi^{(3)} \equiv 0$. Consequently the level-three profile on this open set of c, for all large X, is equivalent to

$$\sum_{\substack{M \le Y \\ M \equiv 3 \pmod 6}} \lambda(M) \lambda(M+1) \lambda(M+3) = o(Y). \tag{8.16}$$

Proof. The first identity is Theorem 8.10(3). Put $N = 6n$. Then $3(2n-1) = N - 3$ and $2(3n-1) = N - 2$, so $\lambda(n) = \lambda(6) \lambda(N) = \lambda(N)$, $\lambda(2n-1) = \lambda(3) \lambda(N-3)$ and $\lambda(3n-1) = \lambda(2) \lambda(N-2)$; since $\lambda(2) \lambda(3) = 1$ the product is $\lambda(N) \lambda(N-2) \lambda(N-3)$. Put $M = N - 3$. □

The logarithmic analogue of (8.16) holds by [12, Theorem 1.1, p. 2]; the ordinary average (8.16) at every scale is not known, and [10] treats only two points [10, pp. 3–4].

8.4. Necessity

Theorem 8.12 (Necessity)**.** *Fix $k \geq 2$ and $a_1, \dots, a_{k-1} \geq 1$. For $r \in \mathbb{Z}_{\geq 0}^{k-1}$ put*

$$P_r(Y) := \sum_{m \leq Y} \lambda(m) \prod_{i=1}^{k-1} \lambda(a_i m + r_i), \qquad \pi_r := \varepsilon_{0,r}\, \mathbf{1}\Big[k'(0,r) = 0\Big], \quad (8.17)$$

with the classes of Definition 8.2 for $(p_i, q_i) = (a_i, 1)$. Suppose that (8.10) *holds for $(p_i, q_i) = (a_i, 1)$ and every $c \in (0,\infty)^{k-1}$. Then $P_r(Y) = \pi_r Y + o(Y)$ for every $r \in \mathbb{Z}_{\geq 0}^{k-1}$. In particular, if the forms $m, a_1 m + r_1, \dots, a_{k-1} m + r_{k-1}$ are pairwise non-proportional, then $P_r(Y) = o(Y)$.*

Proof. Induction on $|r| := \sum_i r_i$. For $r = 0$, $P_0(Y) = \lambda(\prod a_i) \sum_{m \leq Y} \lambda(m)^k$, which is $\pi_0 Y + o(Y)$ by the prime number theorem when k is odd and trivially when k is even. Let $r \neq 0$ and choose $c_i := (r_i + 1)/a_i$. Here $L = 1$, $a = 0$, $n = m$, and $\lfloor e^{u_i} m \rfloor = a_i m + r_i'(m)$ with $r_i'(m) = \lfloor \delta_i m \rfloor$ and $\delta_i = a_i(e^{c_i/X} - 1)$. Since $x < e^x - 1 < x(1+x)$ for $0 < x < 1$,

$$\delta_i m = (r_i + 1)\frac{m}{X}(1 + \eta_i), \qquad 0 < \eta_i < \frac{c_i}{X}.$$

Order of the blocks. For $1 \leq m \leq X$ and $X > (r_i + 1)c_i$ one has $\delta_i m < (r_i + 1)(1 + \eta_i) < r_i + 2$, so $r_i'(m) \leq r_i + 1$, and $r_i'(m) = r_i + 1$ only if $m \geq X/(1+\eta_i) > X - c_i$. Hence, outside the at most $\lceil \max_i c_i \rceil + 1 = O_{a,r}(1)$ values $m > X - \max_i c_i$, every m has $r'(m) \leq r$ componentwise. For each i, $r_i'(m) = r_i$ exactly when $\frac{r_i}{r_i+1}\frac{X}{1+\eta_i} \leq m < \frac{X}{1+\eta_i}$; intersecting over i gives the block of r. Every other block that occurs has $r' \leq r$, $r' \neq r$, hence $|r'| < |r|$, and is an intersection of $k-1$ intervals in m, hence an interval. Consequently

$$I_r(X) = \Big[\alpha_r X,\ X\Big] + O(1), \qquad \alpha_r := \max_i \frac{r_i}{r_i + 1} \in \Big[\tfrac{1}{2}, 1\Big). \quad (8.18)$$

Every other nonempty block has $|r'| < |r|$, is an interval $(\alpha' X, \beta' X] + O(1)$, and by induction contributes $\pi_{r'} |J_{r'}(c)| X + o(X)$. The hypothesis (8.10) and (8.9) leave $\sum_{m \in I_r(X)} \lambda(m) \prod_i \lambda(a_i m + r_i) = \pi_r(1 - \alpha_r)X + o(X)$, that is

$$P_r(Y) - P_r(\alpha_r Y) = \pi_r(1 - \alpha_r)Y + o(Y) \qquad (Y \to \infty). \quad (8.19)$$

Given $\epsilon > 0$ take Y_0 with error $\leq \epsilon Y$ in (8.19) for $Y \geq Y_0$, and J with $\alpha_r^{J+1} Y < Y_0 \leq \alpha_r^J Y$. Telescoping over $Y, \alpha_r Y, \dots, \alpha_r^J Y$ gives $|P_r(Y) - \pi_r(1 -$

$\alpha_r^{J+1})Y| \le \epsilon Y/(1-\alpha_r)+Y_0$, since $|P_r(\alpha_r^{J+1}Y)| \le \alpha_r^{J+1}Y < Y_0$; as $|\pi_r|\alpha_r^{J+1}Y < Y_0$ too, $|P_r(Y)-\pi_r Y| \le \epsilon Y/(1-\alpha_r)+2Y_0$, hence $\limsup_Y |P_r(Y)-\pi_r Y|/Y \le \epsilon/(1-\alpha_r)$. If the forms are pairwise non-proportional, all classes are singletons and $\pi_r = 0$. □

For instance, the level-three profile at $(2,3)$ on $(0,\infty)^2$ implies

$$\sum_{m \le X} \lambda(m)\lambda(2m+1)\lambda(3m+1) = o(X) : \tag{8.20}$$

with $c = (1, \frac{2}{3})$ the blocks are $r = (0,0)$ on the range $t \in [0, \frac{1}{2})$, where $\lambda(m)\lambda(2m)\lambda(3m) = \lambda(m)$, and $r = (1,1)$ on $[\frac{1}{2}, 1]$. Together with Theorem 8.7 and Corollary 8.11, for $q_i = 1$ and $c \in (0,\infty)^{k-1}$ the level-k family is equivalent to k-point Chowla on the tuples it contains (Theorem 8.12); in general, for $k \ge 3$, it is an *exact* reformulation and not a reduction to two points.

Theorem 8.13 (Odd levels, logarithmic average)**.** *Let $k \ge 3$ be odd. For every rational resonance* (8.2) *and every $c \in \mathbb{R}^{k-1}$, $C_X^{(k),\log}(u) \to 0$.*

Proof. The decomposition (8.6) holds termwise with weight $1/n$, $n = Lm+a$; the terms with $m = 0$ contribute $O(1)$, and $\frac{1}{Lm+a} = \frac{1}{Lm} + O_L(m^{-2})$ for $m \ge 1$. A block $(A_X, B_X]$ with $A > 0$ has logarithmic mass $O(1)$; only blocks starting at $A = 0$ need the following input. After a fixed shift making all intercepts positive, each block is $o(\log X)$ by [12, Theorem 1.1, p. 2], which for odd k requires no non-degeneracy condition [12, Remark 1.2, p. 2]. There are $O(1)$ blocks and $H_X \sim \log X$. □

9. Three and more points: the centering step of the bias method

9.1. The bias method for two points

We recall the structure of the proof of the binary corrected Elliott theorem [10, Theorem 1.2, p. 4], Part II of [10], specialised to $f_1 = f_2 = \lambda$. The parameters are [10, (10.1)–(10.2), p. 44]

$$\varepsilon = 10^{-4}, \quad \eta = \frac{\varepsilon}{100}, \quad \kappa_0 = \frac{400}{\eta}, \quad A = e^{2\kappa_0}, \quad c_* = \frac{\varepsilon}{100}, \quad \theta = \frac{1}{4}, \tag{9.1}$$

$$\begin{aligned} &\mathcal{C} = \{p : \ B^{1-\eta} < \log p \le B\}, \qquad \mathcal{Z} = \{p : \ B^{1-\varepsilon} < \log p \le B^{1-\eta}\}, \\ &\mathcal{S} = \mathcal{C} \cup \mathcal{Z}, \qquad P_0 = e^{B^{1-\varepsilon}}, \end{aligned} \tag{9.2}$$

with $v_{\mathcal{C}} = \sum_{p \in \mathcal{C}} 1/p = (\eta + o(1)) \log B$ and $L_0 = \prod_{p \in \mathcal{C}} (1 + A/p) \prod_{p \in \mathcal{Z}} (1 + 1/p)$. An admissible family $\mathcal{D}$ consists of squarefree $\mathcal{S}$-products d with $H < d \le \tau H$, $1 \le H \le e^{C_0 B}$, $\omega(d) \le J = \lceil C_0 \log B \rceil$; $d_{\mathcal{C}}$ is the $\mathcal{C}$-part of d, and the cutoffs are $w_d(x) = \varphi((k_d(x) - v_{\mathcal{C}})/\sqrt{v_{\mathcal{C}}})$ with $k_d(x) = \#\{p \in \mathcal{C} : p \mid x,\ p \nmid d\}$ [10, (10.3)–(10.4), (11.3), pp. 45–47]. The proof has four steps:

1. a persistent bias $|S(N_j)| \ge \gamma N_j$ is assumed;
2. a divisor interval carries mass $\gg L_0/B$ [10, Lemma 11.2, p. 47], and dilation $x = dm$ turns the bias into a lower bound $\gg_\gamma L_0/B$ for a raw sum with ordinary divisibility indicators [10, Lemma 11.3, p. 48];
3. centering the center-band indicators $\mathbf{1}_{p|x} \mapsto \mathbf{1}_{p|x} - \theta/p$ changes the raw sum by $\ll L_0(B^{-1-\varepsilon} + J/P_0)$ [10, Proposition 11.4, p. 49];
4. the centered sum is $\ll L_0 B^{-1-c_*/2}$ by the divisor graph estimate [10, Proposition 10.1, p. 45].

Step (3) is reduced [10, §12.1, pp. 50–51] to the estimate

$$\limsup_{Y \to \infty} \Big| \frac{1}{Y} \sum_{z \le Y} b_1(z) \sum_{M < w \le \tau M} \frac{c_w}{w} \, b_2(z + hw) \Big| \ll B^{-1-\varepsilon}, \tag{9.3}$$

uniformly in coefficients $|c_w| \le 1$ supported on P_0-rough $w \in (M, \tau M]$, $P_0/\tau \le M \le e^{C_0 B}$, and in the twists $b_i(n) = \lambda(n) e(t_i \#\{p \in \mathcal{C} : p \mid n,\ p \nmid u\}/\sqrt{v_{\mathcal{C}}})$. The proof of (9.3) [10, pp. 53–55] uses of b_1, b_2 only their 1-boundedness and the averaged short exponential sum bound [10, Theorem 12.2 and (12.7), pp. 51–52]

$$\limsup_{Y \to \infty} \sup_{\alpha \in \mathbb{R}/\mathbb{Z}} \frac{1}{DY} \int_0^Y \Big| \sum_{y < m \le y + D} b(m) e(\alpha m) \Big| \, dy \ll \frac{\log \log D}{\log D}, \tag{9.4}$$

for one of the two functions, with the supremum outside the integral, together with the rough-number bounds of [10, Lemma 12.3, p. 52]: the sup-norm and fourth-moment bounds (12.9)–(12.10) for $\sum_w c_w w^{-1} e(\alpha h w)$, which rest on the count $\sum_{M < w \le \tau M,\ w\ P_0\text{-rough}} 1/w \ll B^{-1+\varepsilon} \log B$ [10, (12.13), p. 53].

For k points $0 = h_0 < h_1 < \cdots < h_{k-1}$, the same expansion, written out in Lemma 9.9, leads to

$$\mathcal{E}_k := \limsup_{Y \to \infty} \Big| \frac{1}{Y} \sum_{z \le Y} b_0(z) \sum_{M < w \le \tau M} \frac{c_w}{w} \prod_{i=1}^{k-1} b_i(z + h_i w) \Big| \ll B^{-1-\varepsilon}. \tag{9.5}$$

For $f_i = \lambda$ the dilation coefficient of [10, Lemma 11.3] is $a_d = \lambda(d)^k$, and the expansion of [10, §12.1], with $d = uw$ and $x = uz$, produces the coefficients

$c_w^{(u)} = a_{uw}(-\theta)^{\omega(w)}\mathbf{1}[uw \in \mathcal{D}]$ together with the extracted factor $\prod_i \lambda(u) = \lambda(u)^k$. Since w is squarefree, the effective coefficient is

$$\lambda(u)^k\, c_w^{(u)} = \lambda(w)^k(-\theta)^{\omega(w)}\mathbf{1}[uw \in \mathcal{D}] = \begin{cases} \theta^{\omega(w)}\mathbf{1}[uw \in \mathcal{D}], & k \text{ odd}, \\ (-\theta)^{\omega(w)}\mathbf{1}[uw \in \mathcal{D}], & k \text{ even}. \end{cases} \tag{9.6}$$

For odd k every term of the expansion therefore carries a non-negative coefficient, independently of u.

9.2. Barriers

Proposition 9.1 (Polynomial-phase barrier, all $k \geq 3$)**.** *Let $k \geq 3$, $0 = h_0 < h_1 < \cdots < h_{k-1}$, $V := \prod_{0 \leq j < l \leq k-1}(h_l - h_j)$ and $R_i := V \prod_{j \neq i}(h_i - h_j)^{-1} \in \mathbb{Z} \setminus \{0\}$. Let β be irrational and $b_i(m) := e(\beta R_i m^{k-1})$. Then:*

1. *for every integer $D \geq 1$, uniformly in $\alpha \in \mathbb{R}$,*

$$\limsup_{Y \to \infty} \frac{1}{DY} \int_0^Y \Big| \sum_{y < m \leq y+D} b_i(m) e(\alpha m) \Big| dy \leq D^{-1/2}; \tag{9.7}$$

2. *$\prod_{i=0}^{k-1} b_i(z + h_i w) = e(\beta V w^{k-1})$ for all z, w; hence with $c_w := e(-\beta V w^{k-1})\mathbf{1}[w \text{ is } P_0\text{-rough}]$,*

$$\frac{1}{Y} \sum_{z \leq Y} b_0(z) \sum_{M < w \leq \tau M} \frac{c_w}{w} \prod_{i=1}^{k-1} b_i(z + h_i w) = \frac{\lfloor Y \rfloor}{Y} \sum_{\substack{M < w \leq \tau M \\ w \ P_0\text{-rough}}} \frac{1}{w}. \tag{9.8}$$

For $P_0 \leq M \leq P_0^2/\tau$, a range contained in that of (9.3)*, every P_0-rough integer of $(M, \tau M]$ is prime, and the right-hand side of* (9.8) *is $\sum_{M < p \leq \tau M} 1/p \geq (\tau - 1)/(3\tau \log(\tau M)) \gg_\tau B^{-1+\varepsilon}$ for large B, which exceeds $B^{-1-\varepsilon}$ by the factor $B^{2\varepsilon}$. Thus* (9.5) *does not follow from* (9.4)*, the rough support and the size of the coefficients.*

Proof. By Lagrange interpolation, $\sum_i h_i^j \prod_{l \neq i}(h_i - h_l)^{-1}$ equals 0 for $0 \leq j \leq k-2$ and 1 for $j = k-1$. Hence $\sum_i R_i(z + h_i w)^{k-1} = \sum_j \binom{k-1}{j} z^{k-1-j} w^j \sum_i R_i h_i^j = V w^{k-1}$, which is (2). For (1), write $S_y(\alpha)$ for the inner sum; since D is an integer it has exactly D terms $m = n_0 + t$, $1 \leq t \leq D$, $n_0 = \lfloor y \rfloor$. Expanding the square,

$$\frac{1}{Y} \int_0^Y |S_y(\alpha)|^2 dy = \sum_{1 \leq t, t' \leq D} e(\alpha(t - t')) \times \frac{1}{Y} \int_0^Y e\Big(\beta R_i\Big((n_0 + t)^{k-1} - (n_0 + t')^{k-1}\Big)\Big) dy. \tag{9.9}$$

For $t \neq t'$, $n \mapsto \beta R_i((n+t)^{k-1} - (n+t')^{k-1})$ is a polynomial of degree $k-2 \geq 1$ with irrational leading coefficient $\beta R_i(k-1)(t-t')$, so by Weyl's equidistribution theorem its average over $n_0 \leq Y$ tends to 0. There are finitely many pairs, and α enters only through unimodular constants, so the left-hand side of (9.9) tends to D uniformly in α. Cauchy–Schwarz gives $\limsup_Y \frac{1}{Y}\int_0^Y |S_y|\,dy \leq D^{1/2}$. The lower bound for the prime sum is the prime number theorem, with $\log(\tau M) \leq 2B^{1-\varepsilon}$. □

For $f_i = \lambda$ the coefficients (9.6) are not arbitrary: they are $(\pm\theta)^{\omega(w)}$ on the admissible w, and for odd k they are non-negative; Proposition 9.1 does not model them. For odd k the polynomial phases of Proposition 9.1 then no longer produce a constant product. For $k = 3$ the barrier survives with phases of vanishing curvature.

Proposition 9.2 (Barrier for non-negative coefficients, $k = 3$)**.** *Let $0 < h_1 < h_2$, $\rho := h_1h_2(h_1 - h_2)$, $\tau \in (1,2)$, $H, D^* \geq 1$, and let q be a prime with*

$$q \geq \max\Big(2h_2 + 1,\ 7h_2(D^*+1)^3,\ 6|\rho|\tau^2H^2\Big). \tag{9.10}$$

Put $b_0(m) = e((h_1 - h_2)m^2/q)$, $b_1(m) = e(h_2m^2/q)$, $b_2(m) = e(-h_1m^2/q)$. Then:

1. *for integers $1 \leq D \leq D^*$ and every α,*

$$\limsup_{Y\to\infty} \frac{1}{DY}\int_0^Y \Big|\sum_{y<m\leq y+D} b_i(m)e(\alpha m)\Big|dy \leq \frac{3+\log q}{D}; \tag{9.11}$$

2. *for $M \leq H$ and every $c_w \geq 0$ supported on $(M, \tau M]$,*

$$\begin{aligned}\liminf_{Y\to\infty}\ \operatorname{Re}\frac{1}{Y}\sum_{z\leq Y} b_0(z) \sum_{M<w\leq\tau M}\frac{c_w}{w}b_1(z+h_1w)b_2(z+h_2w)\\ \geq\ \frac{1}{2}\sum_{M<w\leq\tau M}\frac{c_w}{w}.\end{aligned} \tag{9.12}$$

Proof. (2) $(h_1 - h_2)z^2 + h_2(z+h_1w)^2 - h_1(z+h_2w)^2 = \rho w^2$, so the product equals $e(\rho w^2/q)$, and $|\rho|w^2/q \leq |\rho|\tau^2H^2/q \leq \frac{1}{6}$ gives $\operatorname{Re} e(\rho w^2/q) \geq \frac{1}{2}$. (1) Let $r \in \{h_1 - h_2, h_2, -h_1\}$, $n_0 = \lfloor y \rfloor$, and $L \leq D+1$ the number of terms. Then $r(n_0+t)^2/q + \alpha(n_0+t) = \text{const} + \vartheta t + rt^2/q$ with $\vartheta = \alpha + 2rn_0/q$, and $\sum_{t\leq L}|e(rt^2/q) - 1| \leq 2\pi h_2L^3/q \leq 1$. So $|S_y| \leq \min(L, \|\vartheta\|^{-1}/2) + 1$. As n_0 runs over a period modulo q, $2rn_0$ runs over all residues ($q \nmid 2r$). The points

$\alpha + j/q$ have spacing $1/q$, at most two of them lie at distance $< 1/q$ from $\mathbb{Z}$, and at most two lie at distance in $[j'/q, (j'+1)/q)$ for each $j' \geq 1$; hence the average of $\min(L, \|\alpha + j/q\|^{-1}/2)$ over j is at most $2L/q + 1 + \log q \leq 2 + \log q$. $\square$

The functions of Proposition 9.2 are q-periodic, so the bound in (1) holds only for windows $D \leq D^*$; this is all that matters. In [10, pp. 54–55] the bound (9.4) is used only at the two window lengths $D_1 \asymp M$ and $D_2 \asymp_h D_1$ fixed before $Y \to \infty$. Choosing $D^* \geq (1 + 2h_2)\lceil H \rceil$ gives $q \asymp_h H^3$, $\log q \ll B$, and for $P_0/\tau \leq D \leq D^*$ the bound in (1) is $O(Be^{-B^{1-\varepsilon}})$, far below $\log\log D/\log D$. So the functions of Proposition 9.2 satisfy (9.4) at every window length used in [10, §12]. Their product along the pattern is $\geq \frac{1}{2}$ in real part, uniformly in u when q is chosen for H, so for odd $k = 3$ no cancellation between different u occurs either. The functions are locally linear phases with slowly varying frequency: on each window the frequency is $\alpha + 2rn_0/q$. Any extension of Step (3) to three points must therefore use information on b_i that excludes such local linear phases, which (9.4) does not.

9.3. Local Gowers norms control the centering error

For a prime N and $g : \mathbb{Z}_N \to \mathbb{C}$ let $\mathbb{E}$ denote averages over $\mathbb{Z}_N$, $\Delta_a g(x) := g(x)\overline{g(x+a)}$, $\|g\|_{U^1} := |\mathbb{E}g|$, and

$$\|g\|_{U^s}^{2^s} := \mathbb{E}_a \, \|\Delta_a g\|_{U^{s-1}}^{2^{s-1}} \qquad (s \geq 2), \tag{9.13}$$

the Gowers norms [8]. In particular $\|g\|_{U^2}^4 = \sum_\xi |\widehat{g}(\xi)|^4 \geq \max_\xi |\widehat{g}(\xi)|^4$, with $\widehat{g}(\xi) = \mathbb{E}_x g(x) e(-x\xi/N)$.

Definition 9.3 (Local Gowers norm)**.** For an integer $D \geq 2$ let N_D be the least prime $> 4D$. For $b : \mathbb{N} \to \mathbb{C}$ and $y \geq 0$ define $G_y : \mathbb{Z}_{N_D} \to \mathbb{C}$ by $G_y(x) = b(\lfloor y \rfloor + x)$ for $x \in \{1, \dots, 2D+1\}$ and $G_y(x) = 0$ for the other residues, and put $\|b\|_{U^s(y;D)} := \|G_y\|_{U^s(\mathbb{Z}_{N_D})}$.

Lemma 9.4 (Generalised von Neumann inequality)**.** *Let $k \geq 2$, $0 = h_0 < h_1 < \cdots < h_{k-1}$, $N > h_{k-1}$ prime, and $g_0, \dots, g_{k-1} : \mathbb{Z}_N \to \mathbb{C}$ with $|g_i| \leq 1$. Then*

$$\left| \mathbb{E}_{z,w \in \mathbb{Z}_N} \prod_{i=0}^{k-1} g_i(z + h_i w) \right| \leq \|g_{k-1}\|_{U^{k-1}}. \tag{9.14}$$

Proof. For $k=2$ the average factorises as $\mathbb{E}g_0\cdot\mathbb{E}g_1$, of modulus $\le \|g_1\|_{U^1}$. For $k\ge 3$, Cauchy–Schwarz in z and $|g_0|\le 1$ give

$$\begin{aligned}\Big|\mathbb{E}_{z,w}\prod_{i=0}^{k-1} g_i(z+h_i w)\Big|^2 &\le \mathbb{E}_z\Big|\mathbb{E}_w\prod_{i=1}^{k-1} g_i(z+h_i w)\Big|^2\\ &= \mathbb{E}_t\,\mathbb{E}_{z,w}\prod_{i=1}^{k-1}(\Delta_{h_i t}g_i)(z+(h_i-h_1)w),\end{aligned} \tag{9.15}$$

after writing $w'=w+t$ and replacing z by $z-h_1 w$. The new shifts are h_i-h_1 for $1\le i\le k-1$; they satisfy $0=h_1-h_1<h_2-h_1<\cdots<h_{k-1}-h_1<N$, so the inductive hypothesis (with $k-1$ functions and the same prime N) applies to them. Since N is prime and $0<h_i<N$, every h_i with $i\ge 1$, and in particular h_{k-1}, is a unit modulo N. By induction the inner average is at most $\|\Delta_{h_{k-1}t}g_{k-1}\|_{U^{k-2}}$ in modulus. Since $t\mapsto h_{k-1}t$ is a bijection of $\mathbb{Z}_N$, Hölder and (9.13) give $\mathbb{E}_t\|\Delta_{h_{k-1}t}g_{k-1}\|_{U^{k-2}}\le(\mathbb{E}_s\|\Delta_s g_{k-1}\|_{U^{k-2}}^{2^{k-2}})^{2^{2-k}}=\|g_{k-1}\|_{U^{k-1}}^2$. □

Proposition 9.5 (Local U^k control of the k-point centering error)**.** *Let $k\ge 3$, $0=h_0<h_1<\cdots<h_{k-1}$, $\tau\in(1,2)$, $M\ge 1$, $\mathcal{J}=(M,\tau M]$, $|c_w|\le 1$ for $w\in\mathcal{J}$, $\varpi:=\sum_{w\in\mathcal{J}}|c_w|/w$, and $D:=\lceil h_{k-1}\tau M\rceil$. There is $C=C(h_{k-1})$ such that for all $b_0,\dots,b_{k-1}:\mathbb{N}\to\mathbb{C}$ with $|b_i|\le 1$ and all $Y\ge 1$,*

$$\begin{aligned}&\frac{1}{Y}\Big|\sum_{z\le Y} b_0(z)\sum_{w\in\mathcal{J}}\frac{c_w}{w}\prod_{i=1}^{k-1} b_i(z+h_i w)\Big|\\ &\qquad\le C\Big(\varpi\log(3M)\Big)^{1/2}\frac{1}{Y}\int_0^Y \|b_{k-1}\|_{U^k(y;D)}\,dy\\ &\qquad\quad+\frac{(2D+1)\varpi}{Y}.\end{aligned} \tag{9.16}$$

No hypothesis on the support of c_w or on $b_0,\dots,b_{k-2}$ is needed.

Proof. Write $g(z)$ for the summand in z, so $|g|\le\varpi$, and for $y\ge 0$ let $\mathcal{W}_y=(y,y+D]\cap\mathbb{Z}$ and $S_y=\sum_{z\in\mathcal{W}_y}g(z)$. Each integer $z\in[D,Y]$ lies in $\mathcal{W}_y$ for a set of $y\in[0,Y]$ of measure D, and every other z for a set of measure in $[0,D]$; hence

$$\Big|\sum_{z\le Y} g(z)\Big|\le\frac{1}{D}\int_0^Y|S_y|\,dy+(2D+1)\varpi. \tag{9.17}$$

Cauchy–Schwarz in w. With $X_{y,w} := \sum_{z\in\mathcal{W}_y} b_0(z)\prod_{i\ge1} b_i(z+h_iw)$ and $|c_w|/w \le 1/M$,

$$|S_y|^2 \le \varpi \sum_{w\in\mathcal{J}} \frac{|c_w|}{w}|X_{y,w}|^2 \le \frac{\varpi}{M}\sum_{w\in\mathcal{J}}|X_{y,w}|^2. \tag{9.18}$$

Differencing. Put $n_0 = \lfloor y\rfloor$, so that $\mathcal{W}_y - n_0 = \{1,\dots,D\}$, and $G_i(x) := b_i(n_0+x)\mathbf{1}_{1\le x\le 2D+1}$ and $F_0(x) := b_0(n_0+x)\mathbf{1}_{n_0+x\in\mathcal{W}_y}$. Writing $z' = z+s$ in $|X_{y,w}|^2$,

$$\sum_{w\in\mathcal{J}}|X_{y,w}|^2 = \sum_{|s|<D} T_s, \qquad T_s := \sum_x \sum_{w\in\mathcal{J}} (\Delta_s F_0)(x)\prod_{i=1}^{k-1}(\Delta_s G_i)(x+h_iw), \tag{9.19}$$

since for $x, x+s \in \mathcal{W}_y - n_0 \subseteq \{1,\dots,D+1\}$ and $w\in\mathcal{J}$ all arguments $x+h_iw$, $x+s+h_iw$ lie in $\{1,\dots,2D+1\}$ (here $h_{k-1}w \le h_{k-1}\tau M \le D$).

Transfer to $\mathbb{Z}_N$. Let $N = N_D \in (4D, 8D]$ (Bertrand). Read F_0, G_i as functions on $\mathbb{Z}_N$ supported on the residues of $\{1,\dots,D+1\}$ and $\{1,\dots,2D+1\}$, and take Δ_s in $\mathbb{Z}_N$. For $|s|<D$, the arguments $x+s$ and $x+h_iw+s$ with $x\le D+1$ lie in $\{2-D,\dots,3D\}$; those outside $\{1,\dots,3D\}$ lie in $\{2-D,\dots,0\}$, whose residues are $\ge N-D+2 > 2D+1$; so Δ_sF_0 is unchanged, and for $w\in\mathcal{J}$ and x in the support of Δ_sF_0 the values $(\Delta_sG_i)(x+h_iw)$ computed in $\mathbb{Z}_N$ equal the integer ones. Thus $T_s = \sum_{x,w\in\mathbb{Z}_N}\mathbf{1}_{\mathcal{J}}(w)(\Delta_sF_0)(x)\prod_{i\ge1}(\Delta_sG_i)(x+h_iw)$, where $\mathcal{J}$ is read in $\mathbb{Z}_N$ ($\tau M < N$). With the normalised transform $\widehat{g}(\xi) = \mathbb{E}_{x\in\mathbb{Z}_N} g(x)e(-x\xi/N)$ fixed after (9.13), Fourier inversion reads $\mathbf{1}_{\mathcal{J}}(w) = \sum_{\xi\in\mathbb{Z}_N}\widehat{\mathbf{1}_{\mathcal{J}}}(\xi)e(\xi w/N)$. For the interval $\mathcal{J}$ of length $|\mathcal{J}|<N$ one has $|\widehat{\mathbf{1}_{\mathcal{J}}}(0)| = |\mathcal{J}|/N \le 1$ and $|\widehat{\mathbf{1}_{\mathcal{J}}}(\xi)| \le \frac{1}{2N\|\xi/N\|}$ for $\xi\neq0$ (geometric sum, $\|\cdot\|$ the distance to $\mathbb{Z}$), so

$$\sum_{\xi\in\mathbb{Z}_N}|\widehat{\mathbf{1}_{\mathcal{J}}}(\xi)| \le 1 + 2\sum_{1\le\xi\le N/2}\frac{1}{2\xi} \le 2+\log N.$$

In T_s the sums over $x,w\in\mathbb{Z}_N$ are unnormalised, so $T_s = N^2\,\mathbb{E}_{x,w\in\mathbb{Z}_N}[\cdots]$; this is the source of the factor N^2 below. Write $e(\xi w/N) = e(\xi\bar{h}_1(x+h_1w)/N)\,e(-\xi\bar{h}_1x/N)$ with $\bar{h}_1h_1\equiv1$. The two linear phases are absorbed into Δ_sF_0 and Δ_sG_1 without changing their sup norms. Since $k\ge3$, the indices 1 and $k-1$ are distinct, so the phase is absorbed into a factor other than the one controlled by the Gowers norm in Lemma 9.4, and that factor,

$\Delta_s G_{k-1}$, is untouched. Applying Lemma 9.4 to $\mathbb{E}_{x,w}$ for each ξ and summing over ξ,

$$|T_s| \le N^2(2+\log N)\,\|\Delta_s G_{k-1}\|_{U^{k-1}(\mathbb{Z}_N)}. \tag{9.20}$$

Summing over s. By Hölder and (9.13),

$$\sum_{|s|<D} \|\Delta_s G_{k-1}\|_{U^{k-1}} \le N^{1-2^{1-k}} \Big(\sum_{s\in\mathbb{Z}_N} \|\Delta_s G_{k-1}\|_{U^{k-1}}^{2^{k-1}} \Big)^{2^{1-k}} = N\,\|G_{k-1}\|_{U^k}^2. \tag{9.21}$$

Combining (9.18)–(9.21), $|S_y| \le (\varpi N^3(2+\log N)/M)^{1/2}\|G_{k-1}\|_{U^k}$. Since $\|G_{k-1}\|_{U^k} = \|b_{k-1}\|_{U^k(y;D)}$, $N \le 8D$ and $D \le 3h_{k-1}M$, one has $N^3/(MD^2) \le 1536h_{k-1}$ and $\log N \ll_{h_{k-1}} \log(3M)$. Insert in (9.17) and divide by Y. □

9.4. A conditional extension of the bias method

Hypothesis 9.6 ($\mathrm{LU}_k(\varrho,K)$: quantitative local Gowers uniformity of λ)**.** Let $k \ge 3$. For every integer $D \ge 2$, every finite set of primes $\mathcal{P}$ and every multiplicative $b:\mathbb{N}\to\mathbb{C}$ with $|b| \le 1$ and $b(p^j) = \lambda(p^j)$ for all $p \notin \mathcal{P}$, $j \ge 1$,

$$\limsup_{Y\to\infty} \frac{1}{Y}\int_0^Y \|b\|_{U^k(y;D)}\,dy \le K\Big(2+\sum_{p\in\mathcal{P}}\frac{1}{p}\Big)^K (\log D)^{-\varrho}. \tag{9.22}$$

Theorem 9.7 (k-point centering under LU_k)**.** *Let $k \ge 3$ and assume $\mathrm{LU}_k(\varrho,K)$ for some K and some $\varrho > 1+3\varepsilon$, with $\varepsilon = 10^{-4}$ as in* (9.1)*. Then* (9.5) *holds for the twists b_i of [10, (12.3), p. 51], any 1-bounded $b_0,\dots,b_{k-2}$, and all coefficients $|c_w| \le 1$ supported on P_0-rough integers of $(M,\tau M]$ with $P_0/\tau \le M \le e^{C_0 B}$, uniformly, for all sufficiently large B.*

Proof. The twist b_{k-1} agrees with λ at all prime powers of primes outside $\mathcal{C}$, and $\sum_{p\in\mathcal{C}} 1/p = v_{\mathcal{C}} \le \log B$. By [10, (12.13), p. 53], $\varpi \ll B^{-1+\varepsilon}\log B$; also $\log(3M) \ll B$ and $\log D \ge \log M \ge B^{1-\varepsilon} - 1$. Proposition 9.5 and (9.22) give

$$\mathcal{E}_k \ll \Big(B^{-1+\varepsilon}(\log B)\,B\Big)^{1/2}(2+\log B)^K B^{-\varrho(1-\varepsilon)} \ll B^{\varepsilon/2}(\log B)^{K+1/2}B^{-\varrho(1-\varepsilon)}, \tag{9.23}$$

Writing the right-hand side as a power of B,

$$B^{\varepsilon/2}(\log B)^{K+1/2}B^{-\varrho(1-\varepsilon)} = B^{-1-\varepsilon}\cdot(\log B)^{K+1/2}\,B^{-\left(\varrho(1-\varepsilon)-1-\frac{3}{2}\varepsilon\right)}. \tag{9.24}$$

If $\varrho > 1+3\varepsilon$, then $\varrho(1-\varepsilon)-1-\frac{3}{2}\varepsilon > (1+3\varepsilon)(1-\varepsilon)-1-\frac{3}{2}\varepsilon = \frac{1}{2}\varepsilon - 3\varepsilon^2 > 0$ for $\varepsilon = 10^{-4}$, so the last factor in (9.24) tends to 0 and $\mathcal{E}_k \ll B^{-1-\varepsilon}$, which is the target (9.5). The boundary term $(2D+1)\varpi/Y$ of (9.16) tends to 0 because D is fixed before $Y\to\infty$. □

For the graph step we formulate the k-linear analogue of [10, Proposition 10.1, p. 45]. With the notation of Subsection 9.1 put

$$g_d^{(k)}(x) := A^{\omega_{\mathcal{C}}(d)}\,\mathbf{1}_{d_{\mathcal{C}}\mid x}\prod_{\substack{p\mid d\\ p\in\mathcal{Z}}}\Big(\mathbf{1}_{p\mid x}-\frac{\theta}{p}\Big)\prod_{i=0}^{k-1}w_d(x+h_id). \tag{9.25}$$

Hypothesis 9.8 ($\mathrm{G}_k(h)$: k-linear divisor graph estimate)**.** Fix $0=h_0<h_1<\cdots<h_{k-1}$, $1<\tau<2$, $C_0\ge 1$, $T>0$. For every sufficiently large B, every admissible family $\mathcal{D}$, all cutoffs w_d as in [10, (10.4)], all $|a_d|\le 1$ and all $F_0,\dots,F_{k-1}:\mathbb{N}\to\mathbb{D}$,

$$\limsup_{X\to\infty}\frac{1}{X}\Big|\sum_{x\le X}\sum_{d\in\mathcal{D}}a_d\prod_{i=0}^{k-1}F_i(x+h_id)\,g_d^{(k)}(x)\Big|\ll L_0B^{-1-c_*/2}, \tag{9.26}$$

with implied constant depending only on h,τ,C_0,T and (9.1).

For $k=2$, (9.26) is [10, Proposition 10.1], proved there through high traces of a self-adjoint graph operator [10, §§13–17, pp. 55–81]; for $k\ge 3$ the natural object is a k-uniform hypergraph with hyperedges $\{x+h_id\}_i$, and the trace method has no direct analogue.

Lemma 9.9 (The k-linear expansion of the centering error)**.** *Let $k\ge 2$, $l\ge 1$, $b\in\mathbb{Z}$, and let B be so large that every prime of $\mathcal{S}$ exceeds l. Let $\mathcal{D}$ be an admissible family with interval $(H,\tau H]$, let $|a_d|\le 1$, let φ be the smooth cutoff of* (9.25) *with $\widehat{\varphi}(t)=\int\varphi(s)e(-ts)\,ds$, and define*

$$R_B(X)=\sum_{\substack{x\le X\\ x\equiv b\ (l)}}\sum_{d\in\mathcal{D}}a_dA^{\omega_{\mathcal{C}}(d)}\mathbf{1}_{d\mid x}\prod_{i=0}^{k-1}\lambda(x+h_id)w_d(x+h_id),$$

and $C_B(X)$ by the same expression with $A^{\omega_{\mathcal{C}}(d)}\mathbf{1}_{d\mid x}\prod_i w_d(x+h_id)$ replaced by $g_d^{(k)}(x)$. Then

$$\limsup_{X\to\infty}\frac{|R_B(X)-C_B(X)|}{X}\le\|\widehat{\varphi}\|_1^k\,L_0\,\sup\mathcal{E}_k, \tag{9.27}$$

where the supremum is over all M with $P_0/\tau\le M\le e^{C_0B}$, all coefficients $|c_w|\le 1$ supported on P_0-rough integers of $(M,\tau M]$, all residue classes $\bar{b}$ modulo l, and all real $t_1,\dots,t_{k-1}$, of the quantity $\mathcal{E}_k$ of (9.5) *with*

$$b_0(z)=\mathbf{1}_{z\equiv\bar{b}\ (l)}\,\lambda(z)\,e\Big(\frac{t_0}{\sqrt{v_{\mathcal{C}}}}\sum_{\substack{p\in\mathcal{C}\\ p\nmid u,\ p\mid z}}1\Big),$$

$$b_i(n) = \lambda(n)\, e\Big(\frac{t_i}{\sqrt{v_{\mathcal{C}}}} \sum_{\substack{p\in\mathcal{C}\\ p\nmid u,\ p\mid n}} 1\Big) \quad (1 \le i \le k-1),$$

$t_0 \in \mathbb{R}$ *and* u *a squarefree* $\mathcal{S}$*-product. Each* b_i *with* $i \ge 1$ *is multiplicative,* 1*-bounded, and agrees with* λ *at every power of a prime outside* $\mathcal{C}$*.*

Proof. Expansion into divisibility indicators. For $d \in \mathcal{D}$ write $d = d_{\mathcal{C}} d_{\mathcal{Z}}$. Since $\mathbf{1}_{p\mid x} - \theta/p$ is affine in $\mathbf{1}_{p\mid x}$ and $d_{\mathcal{Z}}$ is squarefree,

$$\prod_{p\mid d_{\mathcal{Z}}} \Big(\mathbf{1}_{p\mid x} - \frac{\theta}{p}\Big) = \sum_{w\mid d_{\mathcal{Z}}} \frac{(-\theta)^{\omega(w)}}{w}\, \mathbf{1}_{(d_{\mathcal{Z}}/w)\mid x},$$

so $g_d^{(k)}(x) = \sum_{w\mid d_{\mathcal{Z}}} A^{\omega_{\mathcal{C}}(d)} \frac{(-\theta)^{\omega(w)}}{w} \mathbf{1}_{(d/w)\mid x} \prod_i w_d(x+h_i d)$. Multiplying by the outer coefficient a_d and by $\prod_i \lambda(x+h_i d)$, both of which do not depend on w, and summing over x and d,

$$C_B(X) = \sum_{\substack{x\le X\\ x\equiv \bar b\ (l)}} \sum_{d\in\mathcal{D}} a_d \sum_{w\mid d_{\mathcal{Z}}} A^{\omega_{\mathcal{C}}(d)} \frac{(-\theta)^{\omega(w)}}{w} \mathbf{1}_{(d/w)\mid x} \prod_{i=0}^{k-1} \lambda(x+h_i d) w_d(x+h_i d).$$

The term $w = 1$ is exactly the summand of R_B, with the same coefficient a_d. Hence $C_B(X) - R_B(X)$ is the sum of the terms with $w > 1$, each still carrying its coefficient a_d. Write $u = d/w$; then u, w are coprime squarefree $\mathcal{S}$-products, w consists of primes of $\mathcal{Z}$, $\omega_{\mathcal{C}}(u) = \omega_{\mathcal{C}}(d)$, and every prime of $\mathcal{C}$ dividing d divides u. Grouping by u,

$$C_B(X) - R_B(X) = \sum_u A^{\omega_{\mathcal{C}}(u)} \sum_{\substack{x\le X,\ u\mid x\\ x\equiv b\ (l)}} \sum_w \frac{c_w^{(u)}}{w} \prod_{i=0}^{k-1} \lambda(x+h_i uw)\, w_{uw}(x+h_i uw),$$

in which the outer coefficient $a_d = a_{uw}$ has been moved into $c_w^{(u)} := a_{uw}(-\theta)^{\omega(w)}$ if $w > 1$, $uw \in \mathcal{D}$ and $w \mid (uw)_{\mathcal{Z}}$, and $c_w^{(u)} := 0$ otherwise. Thus $|c_w^{(u)}| \le 1$, the support does not depend on x, and u runs over a finite set.

Range of w*.* Put $M = H/u$. If $c_w^{(u)} \ne 0$ then $uw \in (H, \tau H]$, so $w \in (M, \tau M]$; $w > 1$ is a product of primes of $\mathcal{Z}$, all larger than P_0, so w is P_0-rough and $w > P_0$, whence $M > P_0/\tau$; and $M \le H \le e^{C_0 B}$.

Substitution $x = uz$*.* For $u \mid x$ write $x = uz$ with $1 \le z \le Y := X/u$. Every prime of u exceeds l, so $(u, l) = 1$ and $x \equiv b \pmod l$ if and only if $z \equiv \bar b := \bar u b \pmod l$, where $\bar u u \equiv 1 \pmod l$. Since λ is completely multiplicative,

$$\prod_{i=0}^{k-1} \lambda\Big(u(z+h_i w)\Big) = \lambda(u)^k \prod_{i=0}^{k-1} \lambda(z+h_i w),$$

with no coprimality condition. The factor $\lambda(u)^k \in \{\pm 1\}$ depends only on u; it is taken outside the sums over z and w, and it disappears in the absolute value at the end of the proof, since $|\lambda(u)^k| = 1$. The factor $\lambda(z+h_0w) = \lambda(z)$ is kept in b_0. In the cutoff $w_{uw}(n) = \varphi\Big((k_{uw}(n) - v_{\mathcal{C}})/\sqrt{v_{\mathcal{C}}}\Big)$, $k_{uw}(n)$ counts the primes $p \in \mathcal{C}$ with $p \mid n$ and $p \nmid uw$. As w has no prime in $\mathcal{C}$, $p \nmid uw$ means $p \nmid u$, and for such p one has $p \mid u(z+h_iw)$ if and only if $p \mid z + h_iw$. Hence

$$w_{uw}\Big(u(z+h_iw)\Big) = \varphi\Big(\frac{\kappa_u(z+h_iw) - v_{\mathcal{C}}}{\sqrt{v_{\mathcal{C}}}}\Big), \qquad \kappa_u(n) := \#\{p \in \mathcal{C} :\ p \nmid u,\ p \mid n\}.$$

Fourier inversion. Write $\varphi(s) = \int \widehat{\varphi}(t) e(ts)\, dt$ for each of the k cutoffs. The product of the k cutoffs becomes

$$\int_{\mathbb{R}^k} \prod_{i=0}^{k-1} \widehat{\varphi}(t_i)\, e\Big(-\sum_i t_i\sqrt{v_{\mathcal{C}}}\Big) \prod_{i=0}^{k-1} e\Big(\frac{t_i\,\kappa_u(z+h_iw)}{\sqrt{v_{\mathcal{C}}}}\Big)\, dt,$$

and $\lambda(n)e(t_i\kappa_u(n)/\sqrt{v_{\mathcal{C}}})$ is the function $b_i(n)$ of the statement for $i \geq 1$, while $\mathbf{1}_{z\equiv\bar{b}}\lambda(z)e(t_0\kappa_u(z)/\sqrt{v_{\mathcal{C}}}) = b_0(z)$. Since κ_u is additive over coprime factors and takes the same value at p and p^j, each b_i ($i \geq 1$) is multiplicative and 1-bounded and equals λ at powers of primes outside $\mathcal{C}$. Interchanging the finite sums with the integral,

$$\frac{|C_B(X) - R_B(X)|}{X} \leq \sum_u \frac{A^{\omega_{\mathcal{C}}(u)}\,|\lambda(u)^k|}{u} \int_{\mathbb{R}^k} \prod_i |\widehat{\varphi}(t_i)| \Big|\frac{1}{Y}\sum_{z\leq Y} b_0(z) \sum_{M<w\leq \tau M} \frac{c_w^{(u)}}{w} \prod_{i=1}^{k-1} b_i(z+h_iw)\Big|\, dt.$$

Limit. Fix B; the sum over u is finite, and $Y = X/u \to \infty$ with X. Taking $\limsup_{X\to\infty}$, bounding the inner absolute value by its supremum over the parameters listed in the statement, and using $\int |\widehat{\varphi}| = \|\widehat{\varphi}\|_1$ for each of the k variables and $\sum_u A^{\omega_{\mathcal{C}}(u)}/u \leq L_0$ by (9.2) (the sum runs over squarefree $\mathcal{S}$-products) gives (9.27). □

Theorem 9.10 (Conditional k-point Chowla in progressions)**.** *Let $k \geq 3$ and $0 = h_0 < \cdots < h_{k-1}$. Assume $\mathrm{LU}_k(\varrho, K)$ with $\varrho > 1 + 3\varepsilon$ and $\mathrm{G}_k(h)$. Then for all fixed $l \geq 1$ and $b \in \mathbb{Z}$,*

$$\frac{1}{N} \sum_{\substack{m\leq N \\ m\equiv b \pmod{l}}} \prod_{i=0}^{k-1} \lambda(m+h_i) \longrightarrow 0 \qquad (N \to \infty). \tag{9.28}$$

Proof. Suppose the contrary: there are $\gamma \in (0,1]$ and $N_j \to \infty$ with $|S(N_j)| \ge \gamma N_j$, where $S(N)$ is the sum in (9.28). We follow [10, §11] with three changes.

(a) Divisors $\equiv 1 \pmod l$. For large B every prime of $\mathcal{S}$ exceeds l. Under the divisor law of [10, proof of Lemma 11.2] each $p \in \mathcal{S}$ is included independently with probability $A_p/(p+A_p)$, where $A_p = A$ for $p \in \mathcal{C}$ and $A_p = 1$ for $p \in \mathcal{Z}$ (the normalisation of the weights $A^{\omega_{\mathcal{C}}(d)}/d$); for a non-principal character χ modulo l,

$$\begin{aligned} |\mathbb{E}_{\mathrm{div}}\chi(d)| &= \prod_{p\in\mathcal{S}} \Big| \frac{1 + A_p\chi(p)/p}{1 + A_p/p} \Big| \\ &\le \exp\Big(-\sum_{p\in\mathcal{Z}} \frac{1-\operatorname{Re}\chi(p)}{p} + O\Big(\sum_{p>P_0} p^{-2} \Big) \Big) \to 0, \end{aligned} \tag{9.29}$$

because $\sum_{p\in\mathcal{Z}} 1/p = v_{\mathcal{Z}} \to \infty$ while $\sum_{p\in\mathcal{Z}} \chi(p)/p \to 0$ by the prime number theorem in progressions modulo l. Hence $\mathbb{P}_{\mathrm{div}}(d \equiv 1 \pmod l) \to 1/\varphi(l)$. Choosing C_0 so that $\mathbb{P}_{\mathrm{div}}(\omega(d) > J)$ and $\mathbb{P}_{\mathrm{div}}(\log d > C_0 B)$ are each $\le 1/(8\varphi(l))$, the argument of [10, Lemma 11.2] gives an interval $(H, \tau H]$ and a family $\mathcal{D}$ of divisors $d \equiv 1 \pmod l$ with $\sum_{d\in\mathcal{D}} A^{\omega_{\mathcal{C}}(d)}/d \ge c\, L_0/B$, $c = c(l,\tau,C_0) > 0$; the condition $|f_1 f_2(d)| \ge \frac12$ is void for λ.

(b) Raw lower bound with k cutoffs. Put $a_d = \lambda(d)^k$, $X_j = HN_j$, and

$$R_B(X) := \sum_{\substack{x\le X \\ x\equiv b \ (l)}} \sum_{d\in\mathcal{D}} a_d A^{\omega_{\mathcal{C}}(d)} \prod_{i=0}^{k-1} \lambda(x+h_i d) w_d(x+h_i d)\, \mathbf{1}_{d\mid x}. \tag{9.30}$$

For $x = dm$, $d \equiv 1 \pmod l$ gives $x \equiv b \iff m \equiv b \pmod l$, and complete multiplicativity gives $a_d \prod_i \lambda(d(m+h_i)) = \prod_i \lambda(m+h_i)$ with no coprimality condition. As in [10, proof of Lemma 11.3], using that the primes of $\mathcal{C}$ exceed l (so that, by the Chinese remainder theorem, divisibility by them is independent of the class $m \bmod l$), Chebyshev's inequality bounds the proportion of m on which some of the k cutoffs differs from 1 by $16k/T^2$, and the comparison of $X_j/d \in [N_j/\tau, N_j]$ with N_j costs $\tau - 1$. Choosing $\tau - 1 \le \gamma/64$ and $16k/T^2 \le \gamma/64$ gives $\liminf_j |R_B(X_j)|/X_j \ge c_\gamma L_0/B$.

(c) Centering. Let $C_B(X)$ be (9.30) with the factor $A^{\omega_{\mathcal{C}}(d)} \prod_i w_d(x+h_i d)\mathbf{1}_{d\mid x}$ replaced by $g_d^{(k)}(x)$. Lemma 9.9 bounds $\limsup_X |R_B(X) - C_B(X)|/X$ by $\|\widehat{\varphi}\|_1^k L_0 \sup \mathcal{E}_k$, where the supremum is over the parameters listed there. In that family, $b_0, \dots, b_{k-2}$ are 1-bounded and b_{k-1} is

multiplicative, 1-bounded and equal to λ at powers of primes outside $\mathcal{C}$, with $\sum_{p\in\mathcal{C}} 1/p = v_{\mathcal{C}} \le \log B$; the coefficients and M satisfy the hypotheses of Theorem 9.7. That theorem therefore gives $\sup \mathcal{E}_k \ll B^{-1-\varepsilon}$, uniformly in these parameters, and hence $\limsup_X |R_B(X) - C_B(X)|/X \ll L_0 B^{-1-\varepsilon}$, with an implied constant depending on k and φ.

(d) Contradiction. Apply $\mathrm{G}_k(h)$ with $F_0(x) = \mathbf{1}[x \equiv b \pmod l]\lambda(x)$ and $F_i = \lambda$ for $i \ge 1$. It gives $\limsup_X |C_B(X)|/X \ll L_0 B^{-1-c_*/2}$. Hence $c_\gamma L_0/B \le C L_0(B^{-1-c_*/2} + B^{-1-\varepsilon})$, which fails for large B. The order of limits is that of [10, p. 49]: $\gamma, \tau, T, \varphi, C_0$ first, then B fixed while $j \to \infty$, then $B \to \infty$. □

Corollary 9.11 (The k-point family under LU and G)**.** *Let $K_0 \ge 3$. Assume, for every $3 \le k \le K_0$, $\mathrm{LU}_k(\varrho, K)$ with $\varrho > 1 + 3\varepsilon$ and $\mathrm{G}_k(h)$ for every h. Then $\mathrm{Ch}_{k'}$ holds for $3 \le k' \le K_0$, and* (8.10) *holds for every $k \le K_0$, every rational resonance and every c. In particular* (8.16) *holds.*

Proof. Let $\alpha_j m + \beta_j$, $0 \le j < k'$, be pairwise non-proportional, $Q = \prod_j \alpha_j$ and $N = Qm$. Then $(Q/\alpha_j)(\alpha_j m + \beta_j) = N + s_j$ with $s_j = (Q/\alpha_j)\beta_j$, pairwise distinct by non-proportionality, and $\prod_j \lambda(\alpha_j m + \beta_j) = \pm \prod_j \lambda(N + s_j)$. Ordering the s_j and putting $M = N + \min_j s_j$ turns the sum into a sum of $\prod_i \lambda(M + h_i)$ over $M \equiv \min_j s_j \pmod Q$ with $0 = h_0 < \cdots < h_{k'-1}$, which is $o(Y)$ by Theorem 9.10. Theorem 8.7 gives (8.10). □

Remark 9.12 (Status of the two hypotheses). Proposition 9.5, Lemma 9.4 and Propositions 9.1–9.2 are unconditional; Theorem 9.7 is the implication $\mathrm{LU}_k \Rightarrow$(9.5), and Theorem 9.10 is conditional on both hypotheses. At fixed D, LU_k is a statement about averaged 2^k-point correlations of b at shifts $\le 3D$ (expand $\|G_y\|_{U^k}^{2^k}$); it is therefore not a statement obviously weaker than k-point Chowla, and Theorem 9.10 should be read as isolating exactly which uniformity the bias method needs. LU_k concerns windows of fixed length D while $Y \to \infty$, with an ordinary average in y and the rate $(\log D)^{-\varrho}$. Matomäki–Radziwiłł–Tao proved local Fourier uniformity, with the supremum over frequencies inside the integral, for windows $H \ge X^\theta$ with any fixed $\theta > 0$ [9, Theorem 1.2, p. 4]; Tao showed that the logarithmically averaged Chowla conjecture is equivalent to a logarithmically averaged local Gowers uniformity of λ [11]. Windows of fixed length, ordinary averages and a rate, as in (9.22), are not covered by either. Proposition 9.2 shows that, for $k = 3$, some information excluding local linear phases is necessary

for Step (3); Theorem 9.7 shows that the quantitative, uniform local U^3 hypothesis $\mathrm{LU}_3(\varrho, K)$ is sufficient. Whether local U^2 uniformity suffices for the specific non-negative coefficients of (9.6) is open. G_k for $k \geq 3$ is open.

10. Numerical verification

All values below were regenerated from the supplied signed-int8 Liouville prefix using the supplied Python scripts. They are finite checks and are not used as asymptotic rate assumptions.

Experiment 10.1 (Fixed rational resonances)**.** At $X = 2 \cdot 10^6$:

p/q	C_X/X	$\lambda(pq)/q$	$(C_X - X\lambda(pq)/q)/\sqrt{X}$
3/2	0.4999230	0.5000000	−0.109
5/2	0.4995180	0.5000000	−0.682
7/2	0.4996260	0.5000000	−0.529
2/3	0.3326445	0.3333333	−0.974
5/3	0.3324220	0.3333333	−1.289
4/3	−0.3322400	−0.3333333	1.546
5/4	−0.2492080	−0.2500000	1.120

(10.1)

The 2/3 computation omits the unique initial term for which $\lfloor (2/3)n \rfloor = 0$, as required by the definition.

Experiment 10.2 (Natural microscopic profile)**.** For $p/q = 3/2$ and $X = 2 \cdot 10^6$:

c	C_X/X	$\Psi_{3,2}(c)$
−0.5	−0.000265	0
−0.1	−0.000321	0
0.0	0.499923	0.500000
0.3	0.499923	0.500000
0.6	0.499904	0.500000
0.8	0.415884	0.416667
1.2	0.276538	0.277778
2.0	0.166553	0.166667

(10.2)

Experiment 10.3 (Integer endpoint, left interval, and majority)**.** For $m = 2$, $X = 10^6$, $c = 1/2$,

$$Xm(e^{c/X} - 1) = 1.000000250000041666\ldots, \qquad C_X = -999998. \tag{10.3}$$

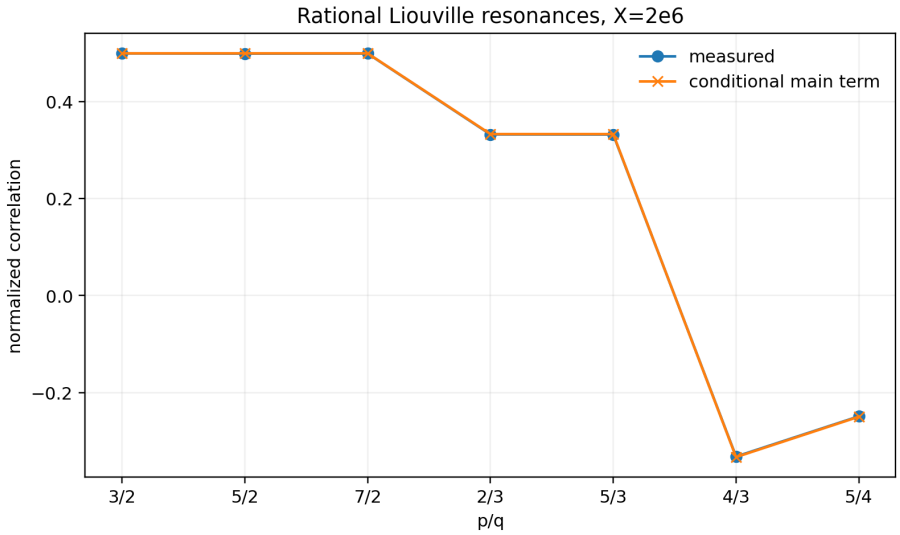


Figure 1: Measured rational resonances at $X = 2 \cdot 10^6$ and the conditional natural main term.

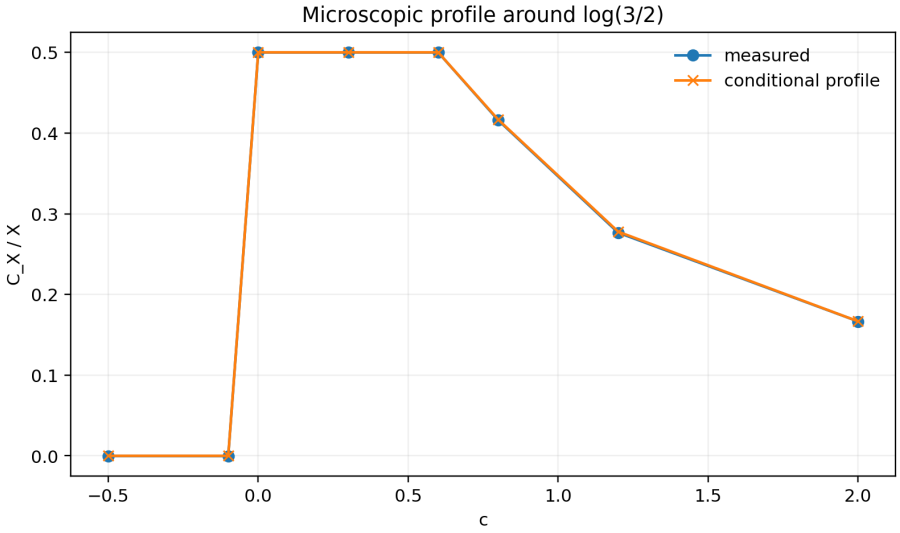


Figure 2: Natural microscopic profile around log(3/2).

For $m = 2$, $X = 10^7$, both $c = -0.01$ and $c = -0.40$ give exactly $C_X = -778$. Two finite checks of Theorem 6.2 are

c	$\lambda(2)C_X/X$	$(2\lvert I_{0,X}\rvert - X)/X$	$2\min(1,(2c)^{-1}) - 1$
0.60	0.833072800	0.666666600	0.666666667
0.90	0.555258200	0.111111000	0.111111111

(10.4)

The middle column is the exact finite-X signed lower bound used in the proof; the last column is its asymptotic limit.

Experiment 10.4 (Logarithmic mesoscopic majority)**.** For $m = 2$, $X = 10^7$ and $c_X = X^\theta$, let B_X denote the exact H_X-normalized harmonic mass of the zero-correction block. The finite bound in the proof is $2B_X - 1$, not its limiting value $1 - 2\theta$. The regenerated values are

θ	c_X	$\lambda(2)C_X^{\log}$	$2B_X - 1$	$1 - 2\theta$
0.00	1.0000	0.958453098	0.916965031	1.000000000
0.20	25.1189	0.765504459	0.530794400	0.600000000
0.40	630.9573	0.573145613	0.144624375	0.200000000
0.45	1412.5375	0.523535345	0.048073466	0.100000000

(10.5)

Thus the finite measured values satisfy the exact finite-X majority bound; the final column is displayed only as the asymptotic lower bound.

11. Conclusions

Reduced rational dilations decompose into one proportional residue class and finitely many non-proportional affine classes. Under fixed affine Chowla this produces the natural microscopic profile $\lambda(pq)\min(1/q, 1/(pc))$ on the positive side and zero on the negative side. Logarithmic weighting behaves differently: the fixed X^{-1} layer collapses to a one-sided step, while the elementary logarithmic majority mechanism has threshold exponent $1/2$ and is effective for every $c_X = X^{\theta+o(1)}$ with $\theta < 1/2$. These unconditional majority statements distinguish integer from non-integer rational resonances without using Chowla, while the conditional and unconditional parts of the argument remain separated explicitly throughout.

With the ordinary two-point theorem of [10] the natural profile is unconditional, with the rate $(\log X)^{-\kappa}$, because its remainder is a finite set

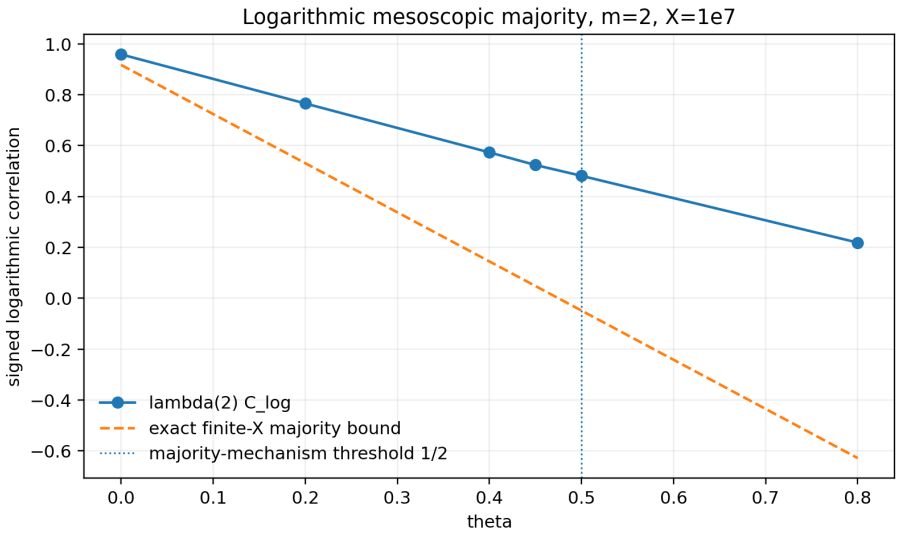


Figure 3: Finite logarithmic mesoscopic check for $m = 2$, $X = 10^7$. The dashed curve is the envelope supplied by the unconditional signed majority bound.

of fixed binary correlations. At level $k \geq 3$ the exact decomposition reduces every block to a k'-point correlation with $k' \equiv k \pmod 2$; blocks with $k' \leq 2$ are unconditional, and the left-hand identity shows that blocks with $k' \geq 3$ are genuine, for every large X: level three at $(2, 3)$ is the ordinary sum $\sum_{M \equiv 3\,(6)} \lambda(M)\lambda(M+1)\lambda(M+3)$. In the bias method the centering step is controlled by local U^k norms, and cannot be controlled by the averaged short exponential sums used for two points; the graph step requires a k-linear divisor-graph estimate. Both inputs are isolated as hypotheses.

12. Data and reproducibility

The numerical calculations use only `liouville_int8_N21000000.bin`, containing the first 21,000,000 Liouville values as signed int8 with byte $n-1$ equal to $\lambda(n)$. The binary is deterministically reproducible from $\lambda(n) = (-1)^{\Omega(n)}$ using the supplied generator. The repository contains every Python script used for the reported checks, all corresponding stdout captures, CSV and text outputs, the three manuscript figures as PNG files, and SHA-256 inventories for the submission files and the complete repository.

13. Use of AI tools

AI tools were used during drafting, code development, and consistency checking. All mathematical claims retained in the manuscript were reviewed by the author. All reported numerical values were regenerated from the supplied Liouville binary by the supplied Python scripts and checked against the identities stated in the manuscript. The author verified the resulting computations and remains responsible for the manuscript and its numerical content.

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